INTRODUCTION

This article outlines the field engineering procedures to be followed when inspecting the MS 6001B gas turbine load gear.

This article covers load gear casing rigging and disassembly, pinion and bull gear shaft inspection, journal bearing inspection, alignment fixturing, and re-assembly of the load gear.

DESIGN INFORMATION

Load Gears provided with MS 6001B Turbine Generator Sets are an up-over, double helical, high speed design which places the high speed pinion gear shaft (that connects to the MS 6001B Gas Turbine) above the lower speed bull gear that connects to the generator.

This gear type is known as a S674-A Load Gear (Philadelphia Drawing 03031-2457.) They are currently supplied by Philadelphia Gear, however this procedure also applies to Westech Load Gears that have been provided for some MS 6001 Gas Turbine units. (See Figs. 1 and 2).

INSPECTION RECOMMENDATIONS

The load gear inspection should be performed during a gas turbine maintenance inspection. A gear tooth contact check should be conducted prior to disassembly of the load gear/generator coupling. Refer to EM 5210 for the GE approved procedure.

The load gear vendor must be requested to be at site during the bearing inspection if the gear is still in warranty. The load gear vendor should be consulted if any abnormal conditions are encountered during an out-of-warranty bearing inspection.

REFERENCE DOCUMENTS

No.	Description
HSM–175	Philadelphia Load Gear Service Manuel
TH 511, TH 514	Internal Alignment of Ship Propulsion Gears – Procedures for Tooth Contact
EM 5210	MS 61B Load Gear – Contact Check Field Inspection
EM 5315	Installation Procedure – MS 6001B Gas Turbine-Generator Unit
EM 5316	MS 61B (Lucas Aero Space) Load Coupling Installation.
ML 0411	Field Alignment Instructions

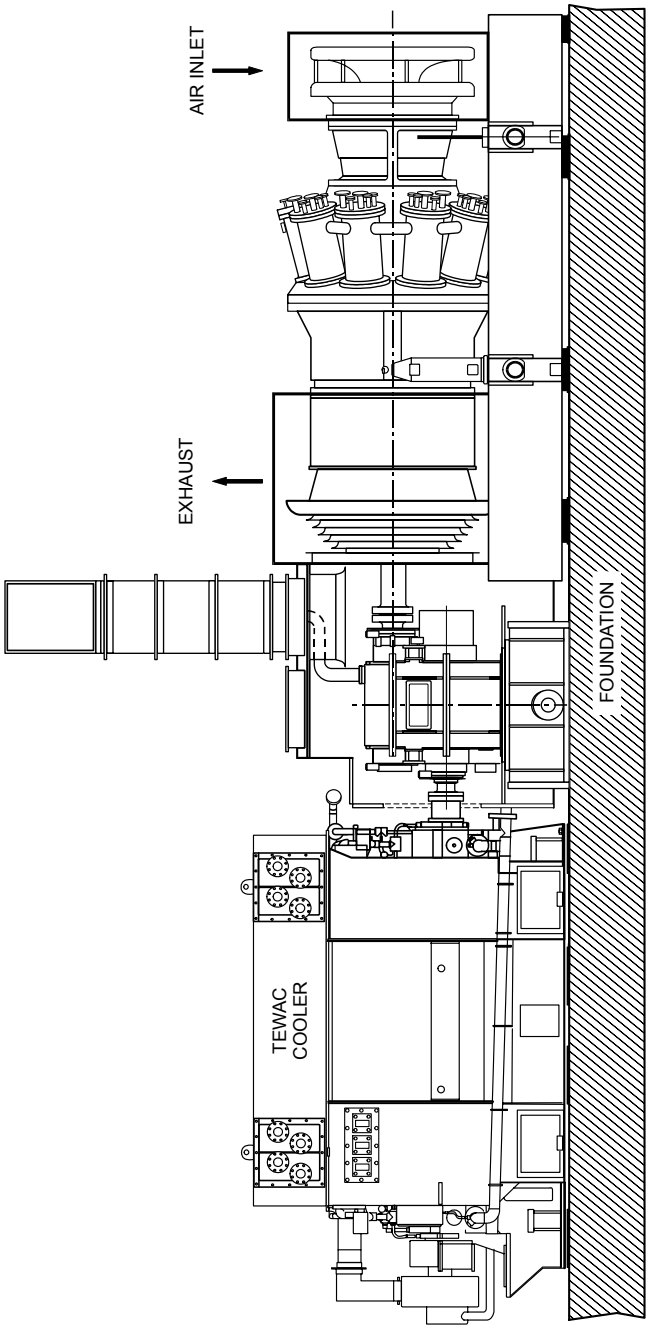
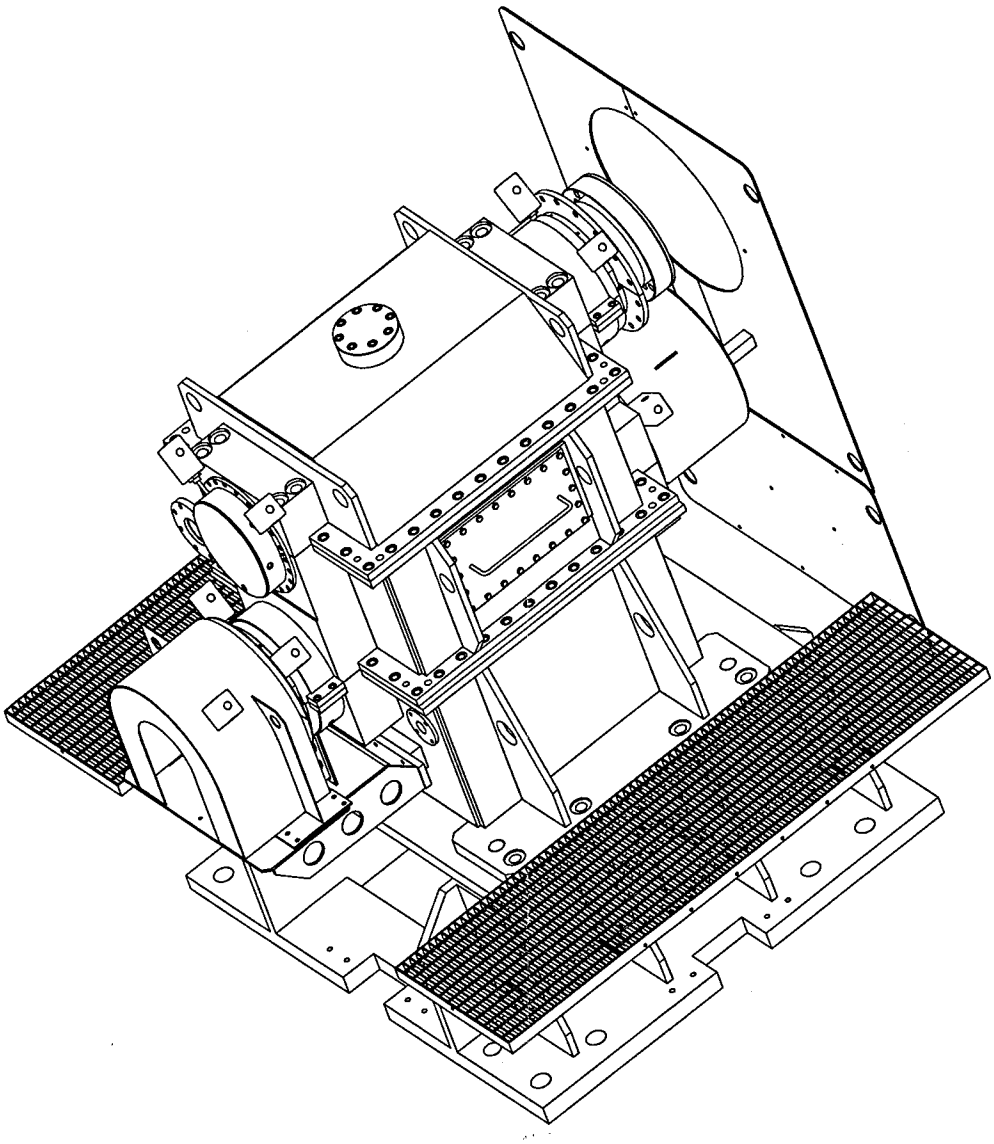


Figure 1.
MS 6001B Gas Turbine w/DLN-1 Combustion, S 674 Load Gear & 6A6 Generator – Typical



GT61B008

Figure 2
Philadelphia S–674 Load Gear — Typical Layout (Plan View)

SPECIAL TOOLS AND FIXTURES

The special tools and fixtures furnished by General Electric during the initial installation will be required for load gear inspection. These include the field fabricated gear turning tool (or coupling wrench), the alignment centering ring provided by the manufacturer (when re-alignment is necessary), and the field fabricated load gear clamp fixtures. Other tools necessary for the inspection include the alignment fixture (provided in ML A033), and a Master Level (Model 98).

STANDARD TOOLING REQUIREMENTS

Standard tooling will be required to remove bolting on the gear box flanges. The rigging equipment required to disassemble and remove upper cover, pinion gear shaft and upper gear box housing include; cables, nylon straps, shackles, 10 ton chain fall, two 1-1/2 ton come-a-longs, torque wrench (with a range 0–900 ft. lb.), and a 1-1/2 ton port-a-power.

METHOD OF INSPECTION

The following is the recommended method for inspection of the MS 61B Load Gear main bearings and journals at both ends of the pinion and bull gear shafts.

1. Before disassembly of the couplings start, insure that the turbine-generator and auxiliaries are properly locked out and tagged.
2. Remove the inspection cover from the left side of the gear box and perform a gear tooth contact check inspection. (Refer to EM 5210 for recommended procedures.) Some newer gear cases may have the inspection cover on the right hand side. (See Fig. 3).

Gear Enclosure Disassembly

The gear enclosure roof must be removed in order to disassemble and inspect the load gear shaft journals and main bearings.

1. Disassemble all of the internal and external electrical wiring, fire protection (CO_2) piping and instrumentation connections from enclosure roof and vent duct areas. (See Fig. 4).

   Note: On older enclosures, the lube oil feed and drain piping will have to be disassembled so that the complete enclosure can be removed.

2. Disassemble and remove the air inlet and air discharge vent duct–rain hood assemblies. (See Figs. 5 and 6, Detail A-A).

   Note: Do not remove the damper assemblies from the enclosure roof.

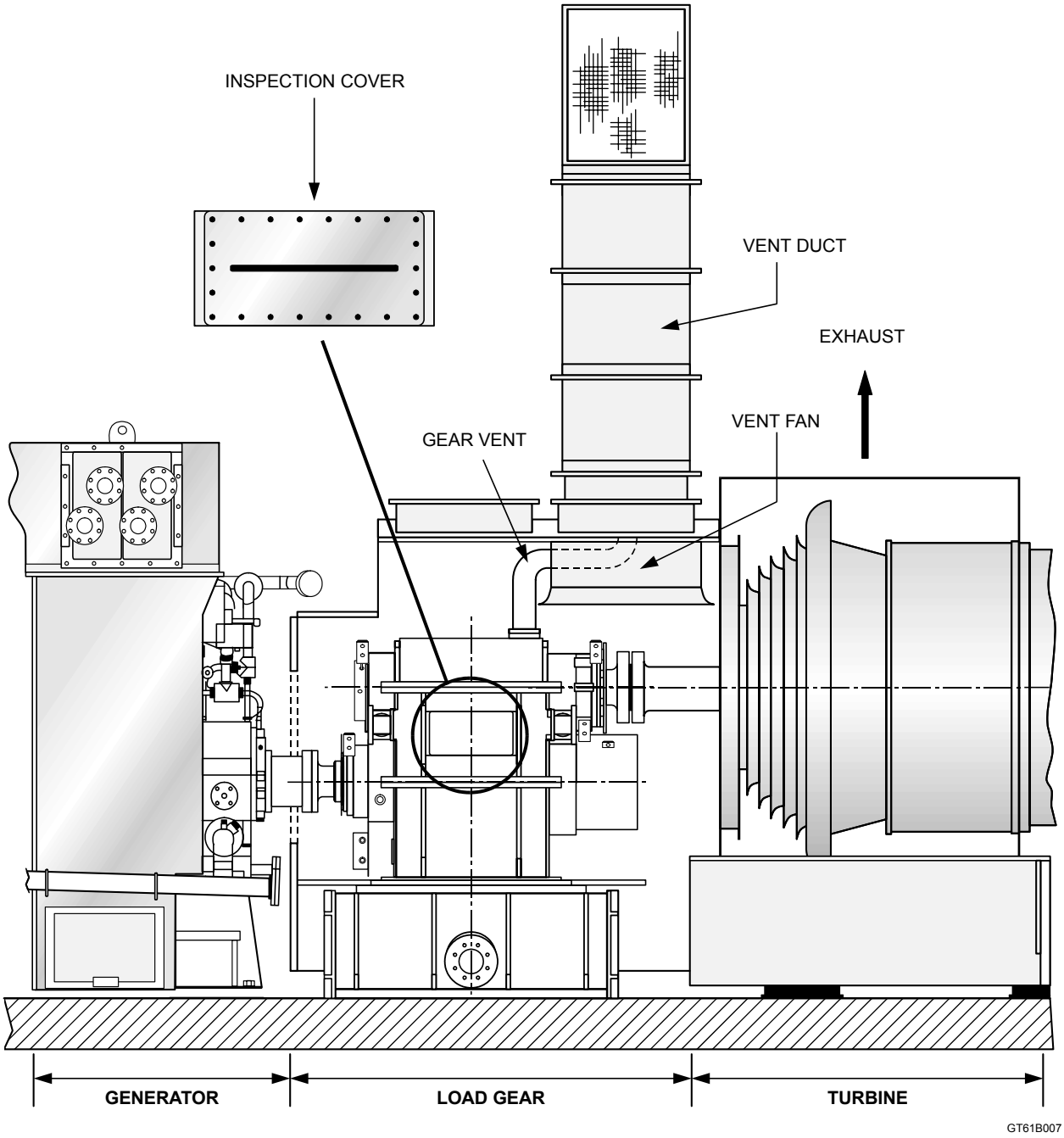


Figure 3. Philadelphia S–674 Load Gear

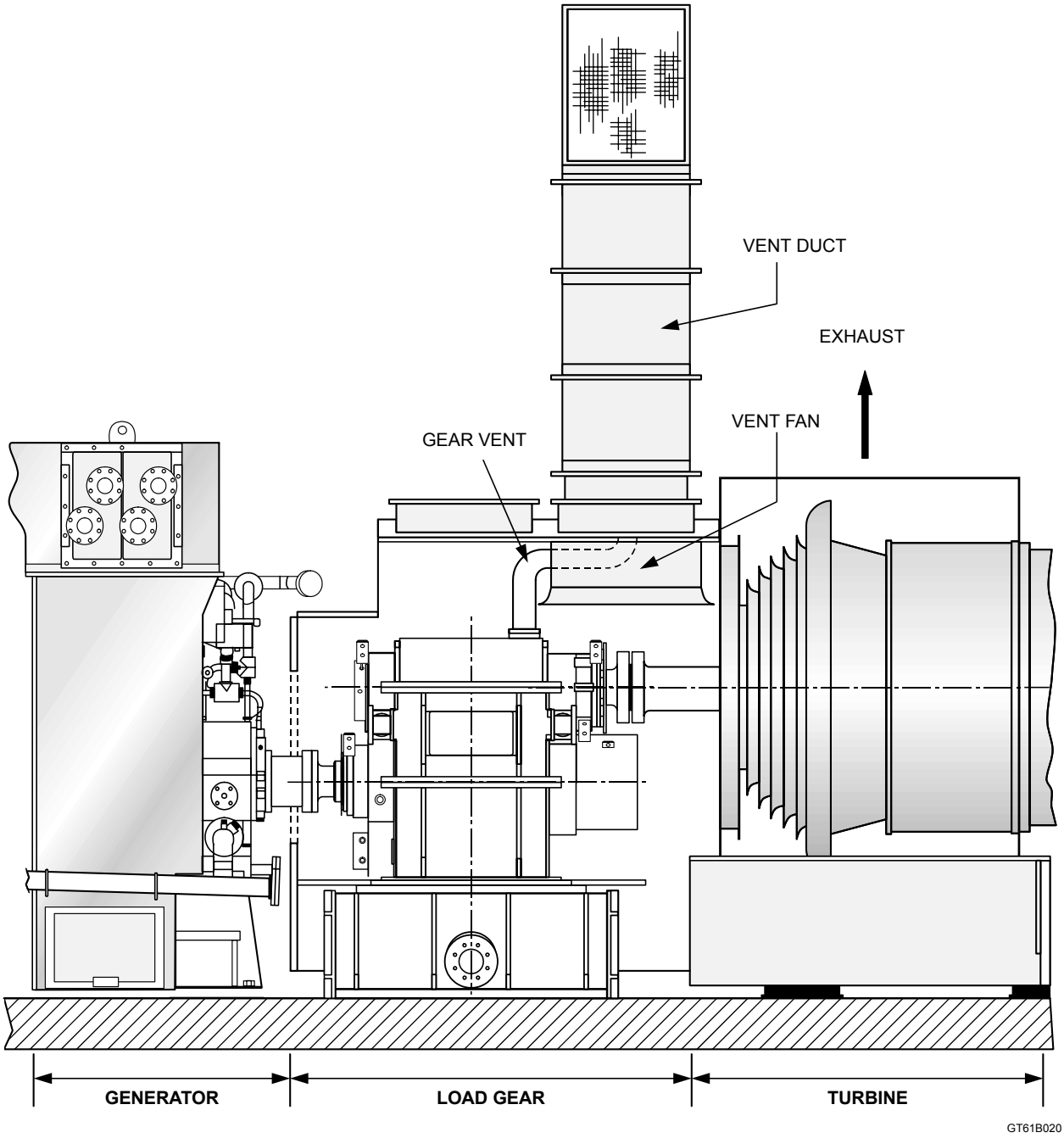


Figure 4. Roof Enclosure Vent Piping Connections

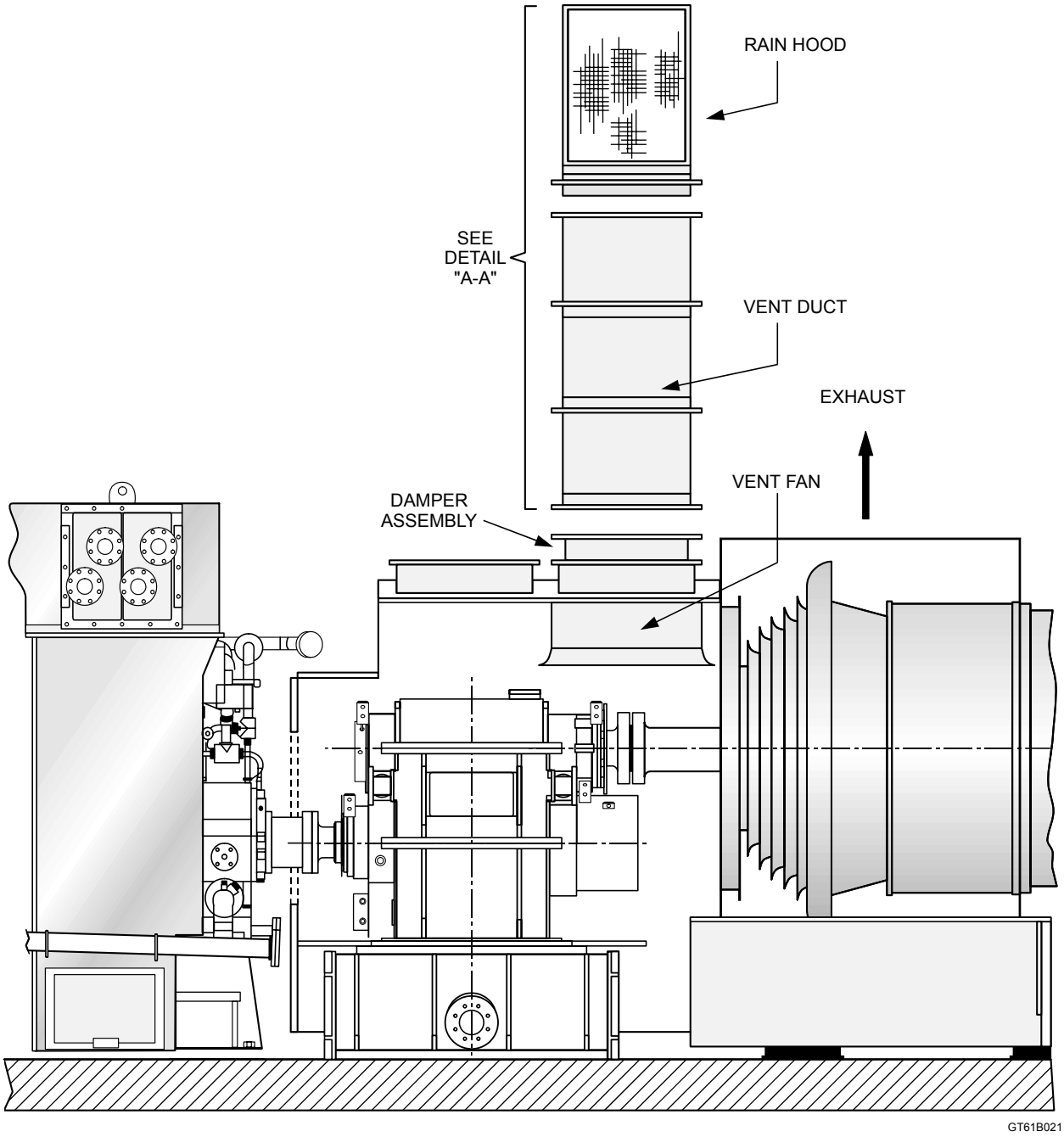
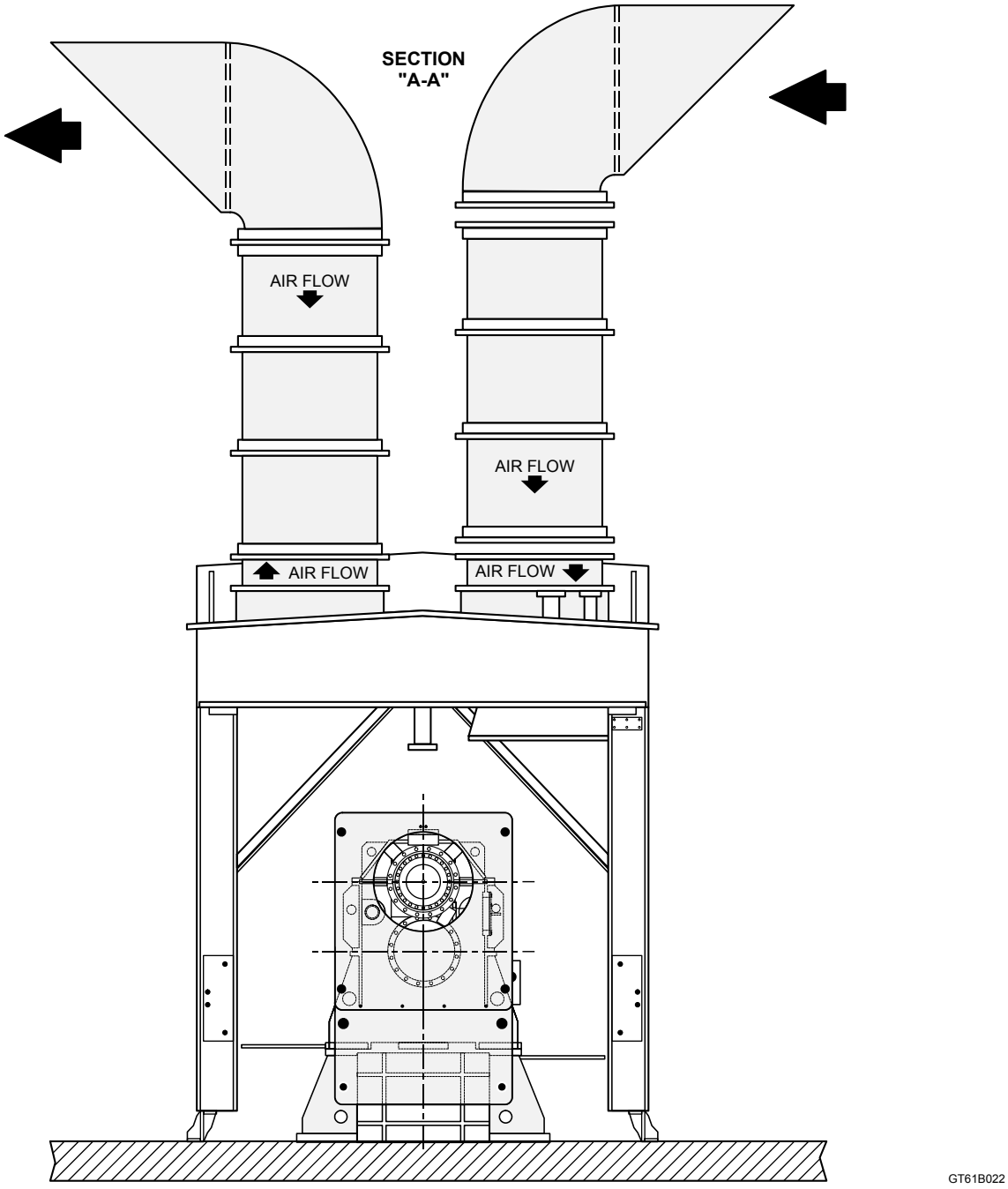


Figure 5. Air Inlet and Vent Duct Assembly



Note: View is from the turbine-end looking towards generator.

Figure 6. Air Inlet, Aid Discharge Vent Duct–Rain Hood Assemble Detail "A-A"

3. Unbolt and remove the enclosure roof. Do not remove the vent fan or the damper assemblies. (See Fig. 7).

Load Gear Disassembly and Alignment Readings

Remove the turbine load coupling. Then remove the generator coupling bolts and jack generator coupling apart. When the coupling is separated jack the generator rotor aft (toward the collector end) 0.750 inches. (See Fig. 8).

Note: Detailed instructions covering the removal of the couplings are provided in EM 5316–MS 61B Load Coupling Installation, EM 5315–Installation Procedure for MS 6001B Gas Turbine-Generator Unit, and ML 0411–Field Alignment Instructions.

Generator Coupling Readings

1. Remove the lube oil seal from the load (bull) gear shaft (generator end), then jack the shaft upward enough to insert the factory supplied alignment centering ring. (See Figs. 9 and 10).

WARNING: The centering ring must be removed after alignment is complete and prior to turbine operation.

Note: The centering ring is supplied by the vendor and turned over to the customer at the end of the installation phase. It is required for all generator alignment work.

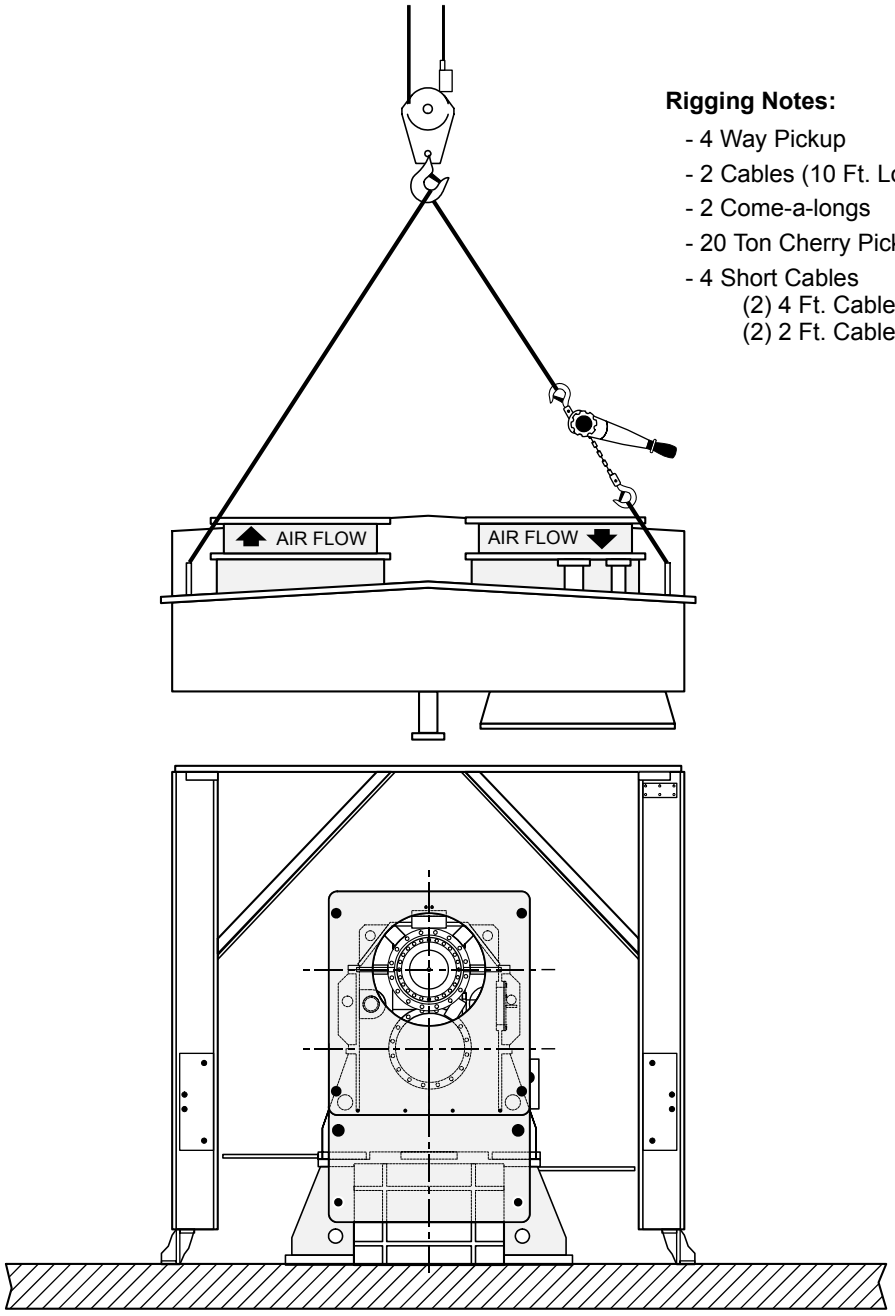
2. Assemble a dial indicator strap chain around the bull gear shaft hub (generator end), then attach two (2) dial indicators to the chain mount, and measure the generator coupling face and rim readings. (See Fig. 11).

3. Assemble a dial indicator on the load gear (generator end), and measure the float of the bull gear shaft to obtain a correct face reading. (See Fig. 11).

4. Install a shaft clamp fixture around the pinion gear shaft (turbine end) where it exits the gear case. Leave a gap of approximately 0.002 inches between gear case and clamp. (See Fig. 12).

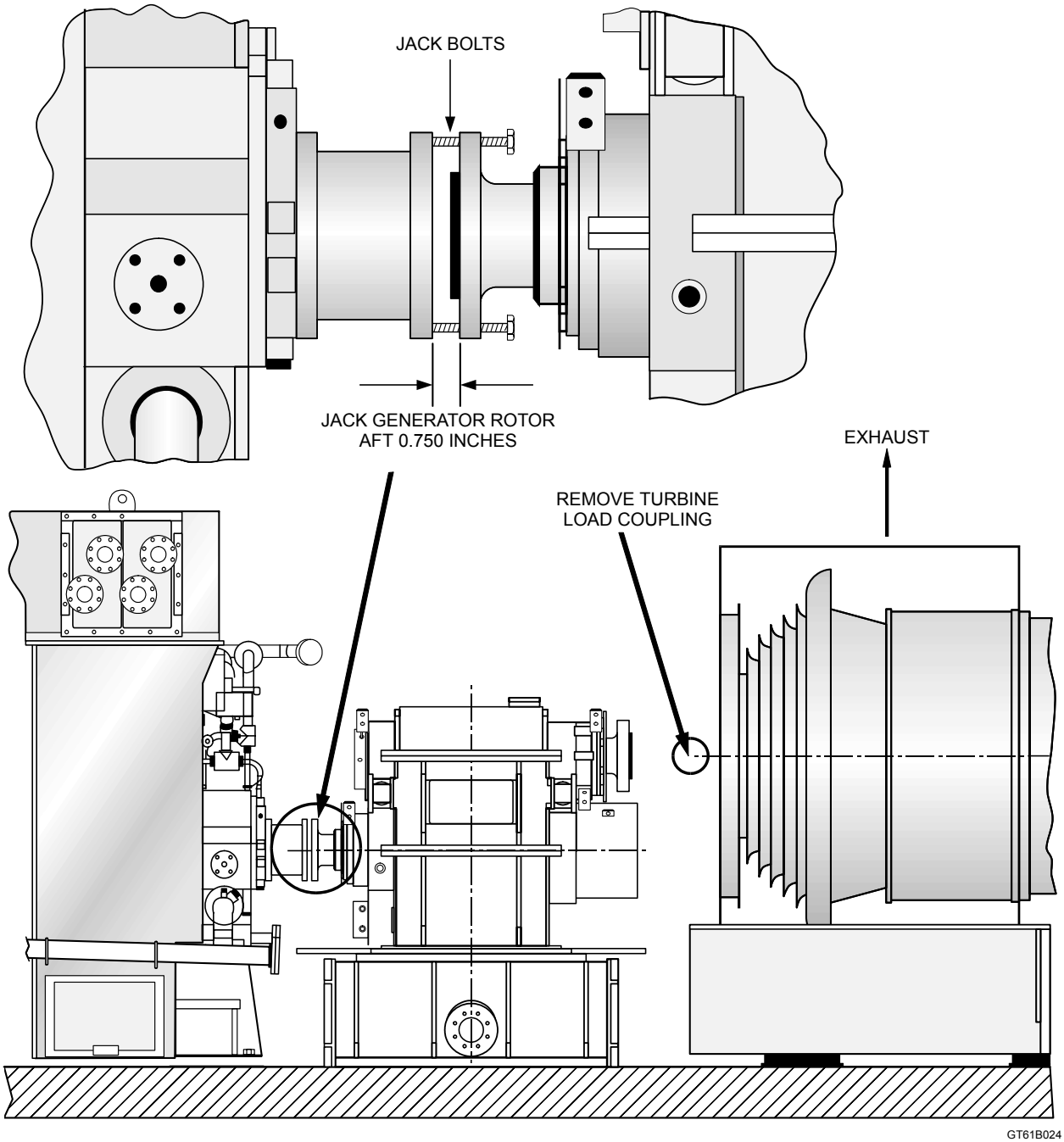
5. Install a shaft clamp fixture around the bull gear shaft (generator end) where it exits the gear case. Leave a gap of approximately 0.002 inches between gear case and clamp. These fixtures will help control most of the gear float during rotation. (See Fig. 12).

Note: These clamp fixtures should have been fabricated on-site during the installation phase for the initial alignment of the gas turbine and generator to the load gear. These special tools are normally turned over to the customer at the end of the installation.



Note: View is from the turbine-end looking towards generator.

Figure 7. Load Gear Roof Enclosure Rigging — Typical Arrangement



Note: Remove jack bolts after coupling separation.

Figure 8. Load Coupling Disassembly

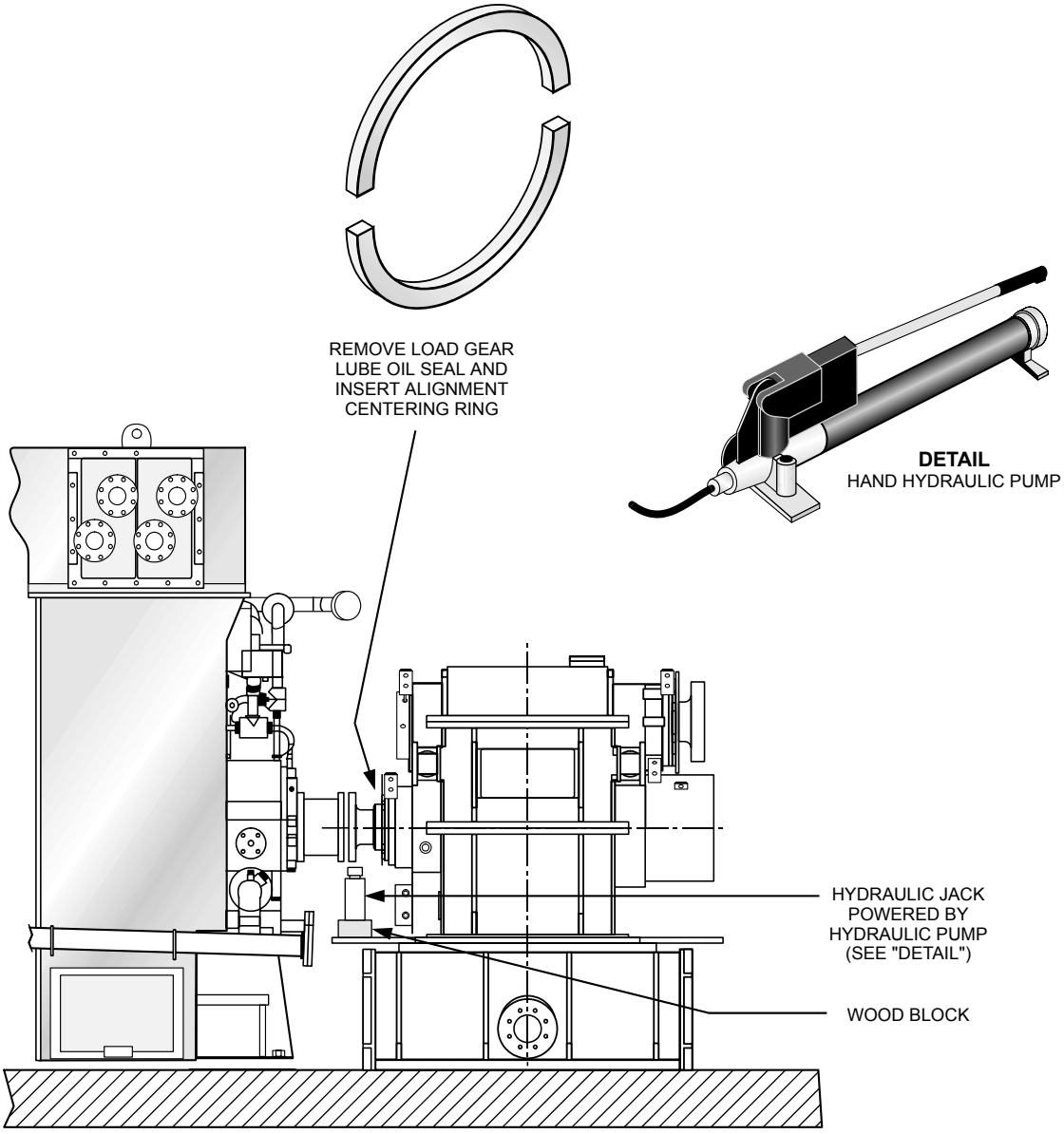


Figure 9. Centering Ring Installation Arrangement

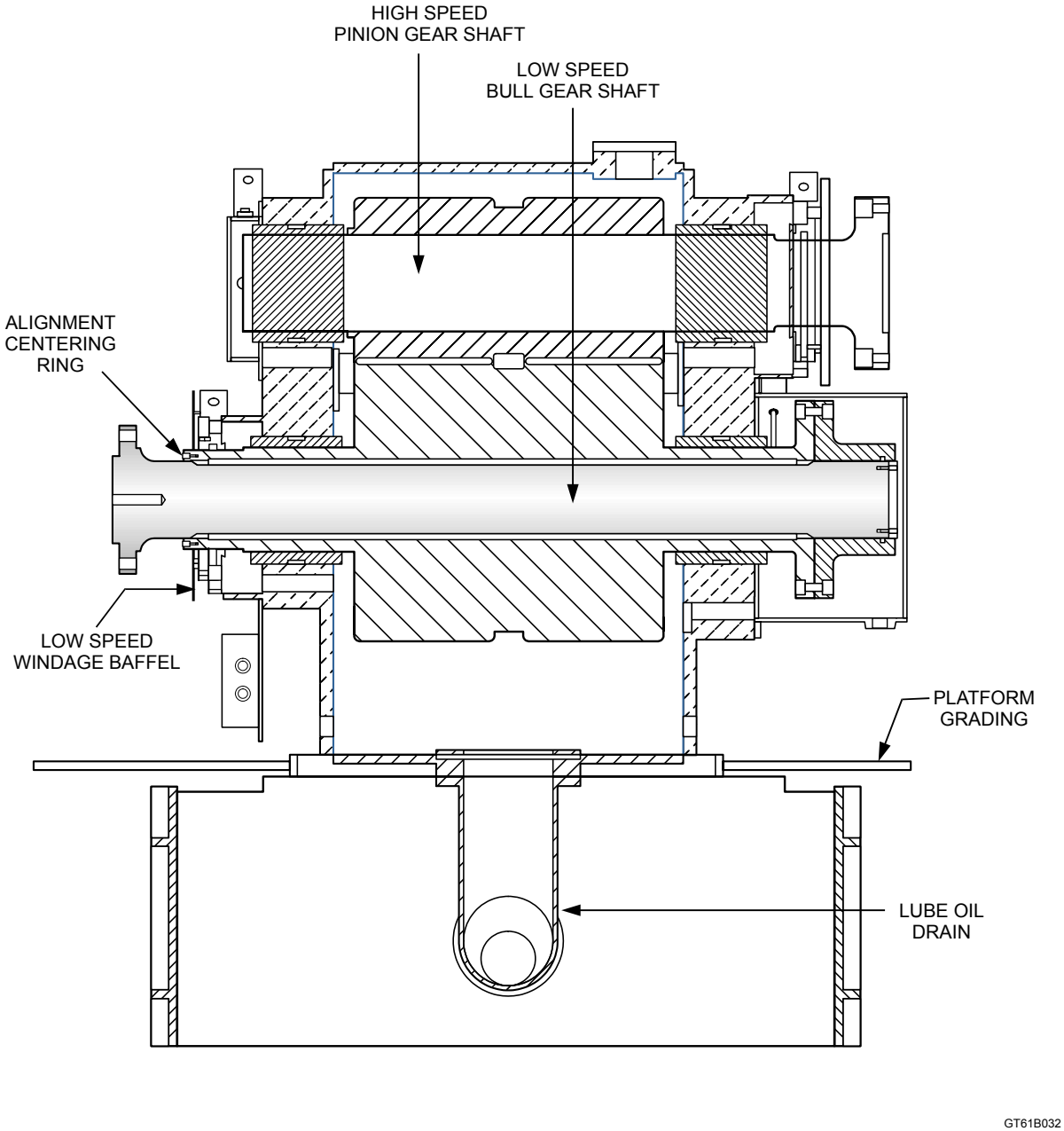


Figure 10. Centering Ring Installation Arrangement Detail

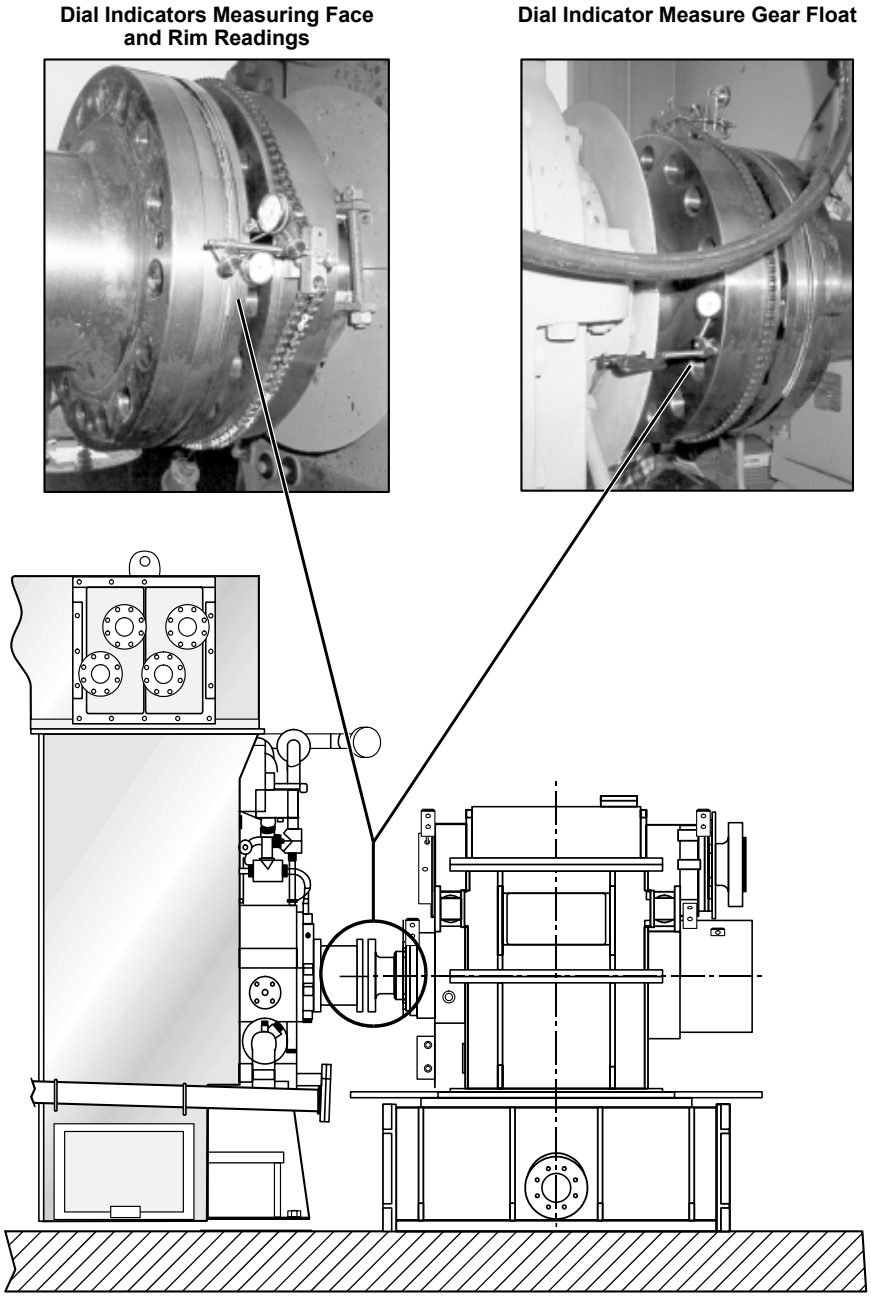


Figure 11. Indicator Arrangement for Rim Readings

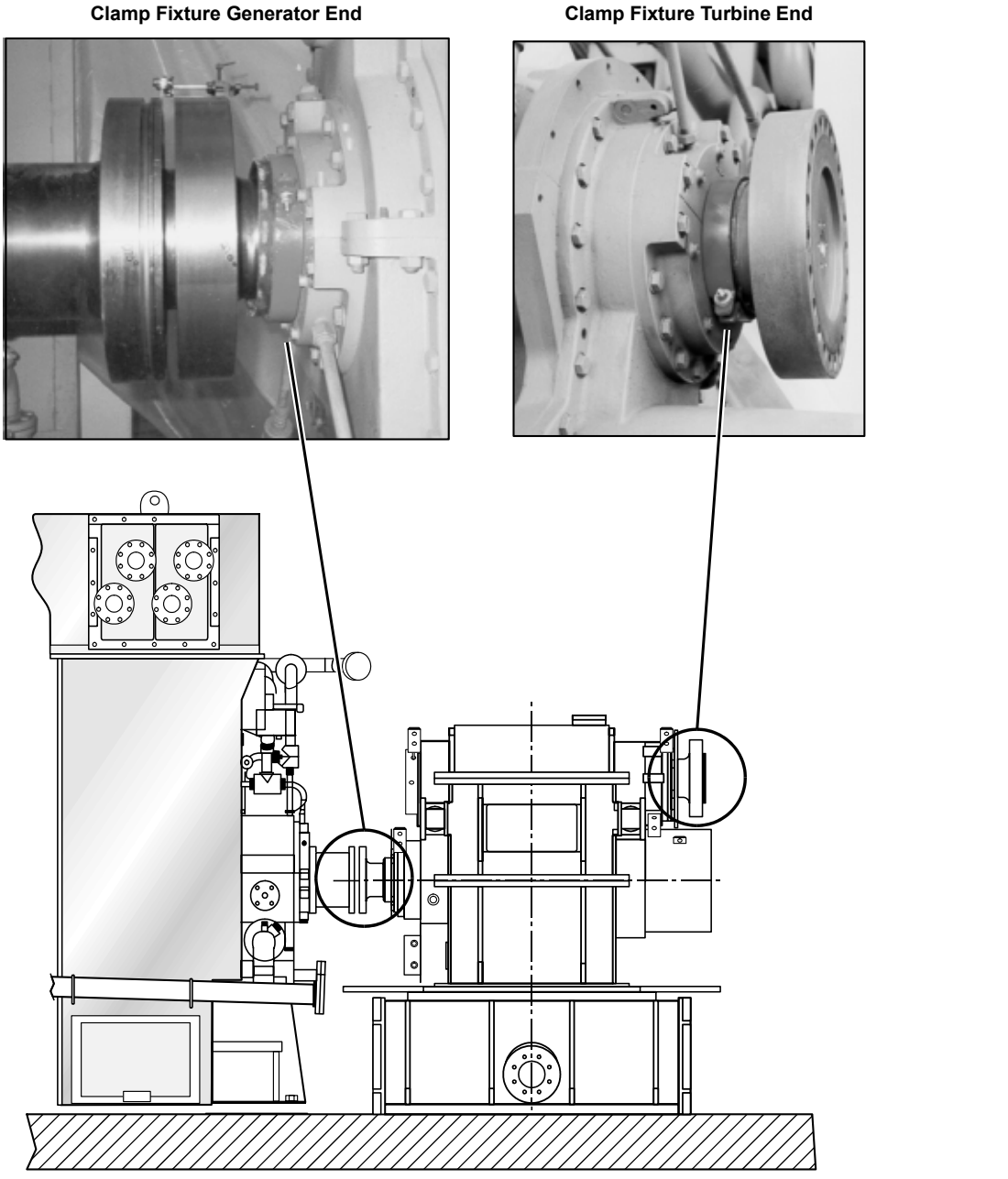


Figure 12. Bull and Pinion Gear Shaft Fixture Installation

6. Use the coupling wrench (fabricated during the installation phase) to rotate the load gear 360 degrees. Record the "as-found" face and rim reading. (See Fig. 13).

Gas Turbine Coupling Readings

1. Assemble the alignment fixture to load gear pinion gear shaft. (See Fig. 14).
2. Attach two dial indicators to alignment fixture measure face and rim readings on the turbine rotor. (See Fig. 14).

Note: Refer to Generator Alignment section steps 3 thru 6. Record the "as-found" face and rim readings.

Load Gear Disassembly

The following is the recommended method of disassembling the MS 61B load gear to inspect the pinion and bull gear shafts journals and main bearings.

1. Remove bolts and dowels from gear box upper cover horizontal joint, install four guide pins, then use four (4) jack bolts to lift the cover up approximately 1.000 inch, then rig, level and slowly remove upper cover. (See Fig. 15).
2. Remove the upper half of both main journal bearings from the pinion gear shaft. Inspect for damage, clean each with a soft rag and mark (“pinion bearing turbine end” and “pinion bearing–generator end”.) (See Fig. 16).

Note: Draw a match mark across one end the pinion and bull gear teeth with a Magic Marker.

3. Attach two dial indicators; one on the turbine end and one on the generator end to read the upward movement of the gear shaft. (See Fig. 16).
4. Assemble the following rigging equipment to remove the pinion gear: (See Fig. 17).

   – Chain fall attached to the overhead crane hook
   – Two short cables
   – Two come-a-longs
   – Two hardwood blocks (one on either side of the gear teeth)
   – Two nylon straps
   – Master level (Series 98)

Note: The required crew normally consists of four (4) millwrights: One to operate the come-a-longs and chain fall, two to monitor the bearings, and one to monitor the gear teeth through the inspection cover hole as they disengage.

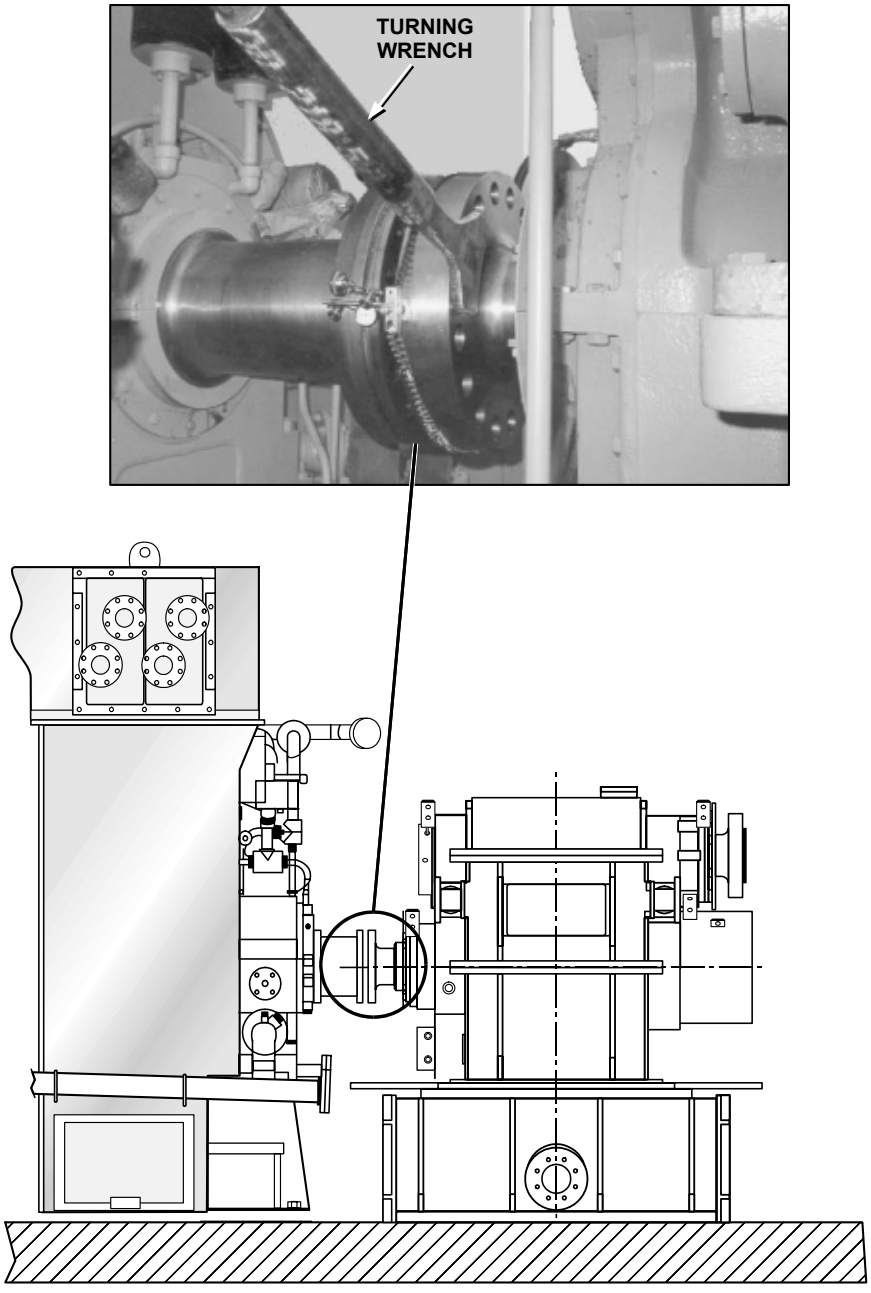


Figure 13. Fabricated Coupling Wrench for Turning Load Gear

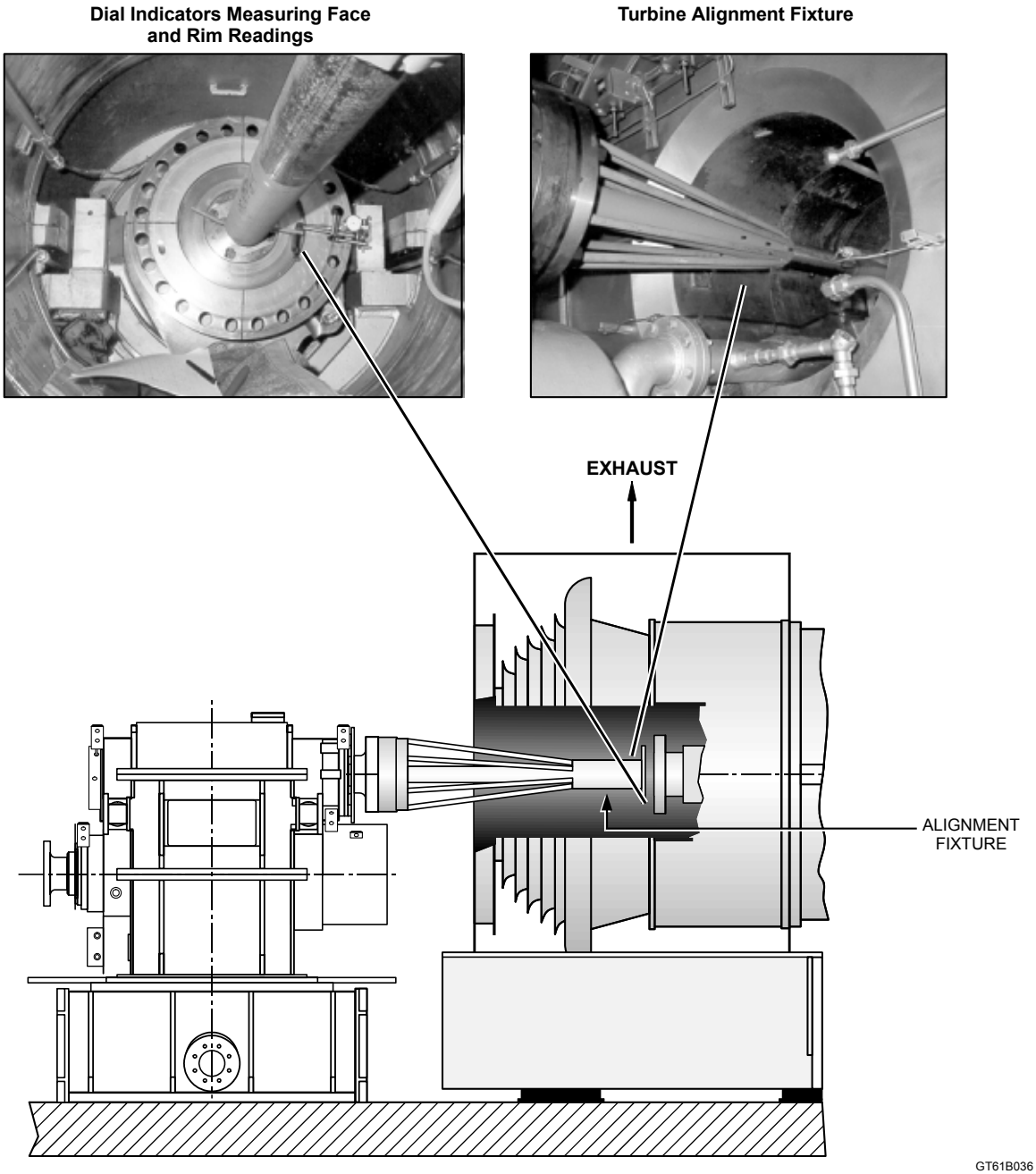
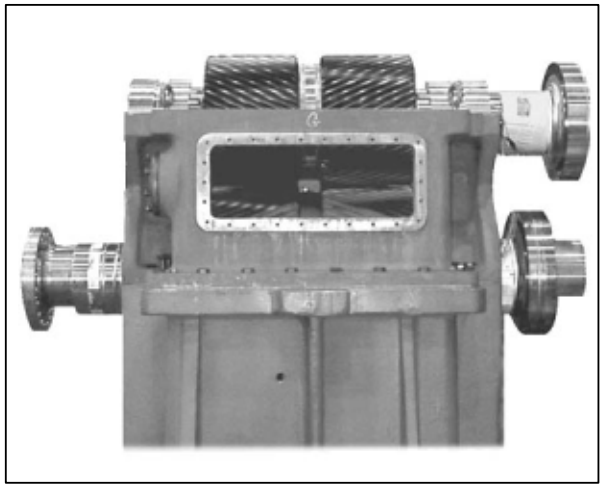
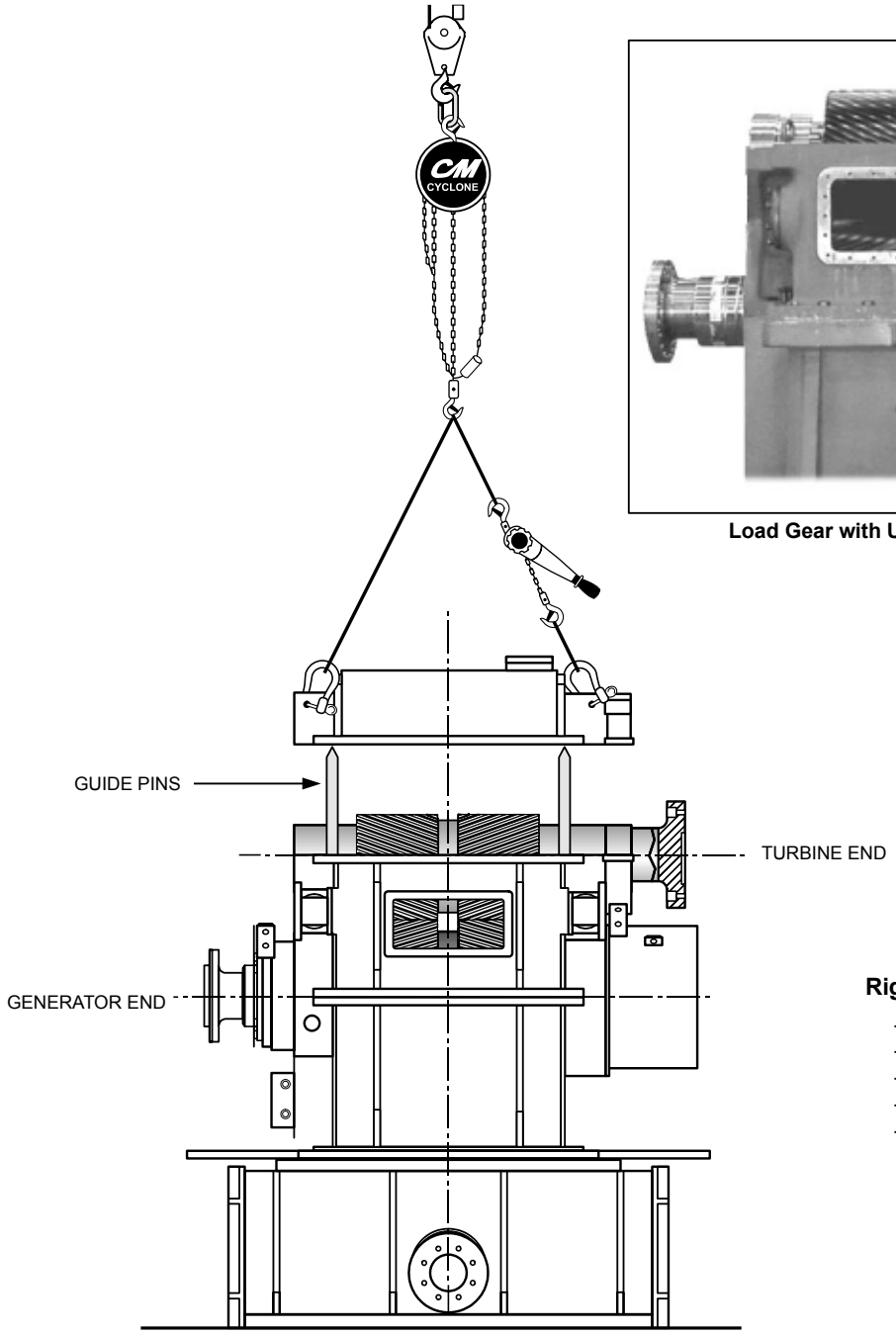


Figure 14. Turbine Alignment Fixture with Dial Indicator Assembly



Load Gear with Upper Cover Removed



Rigging Notes:

- 4 Guide Pins
- 4 Way Pickup
- 2 - 8 ft. Cables
- 2 Come-a-longs
- 4 Short Cables

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Figure 15. Load Gear Upper Cover Rigging

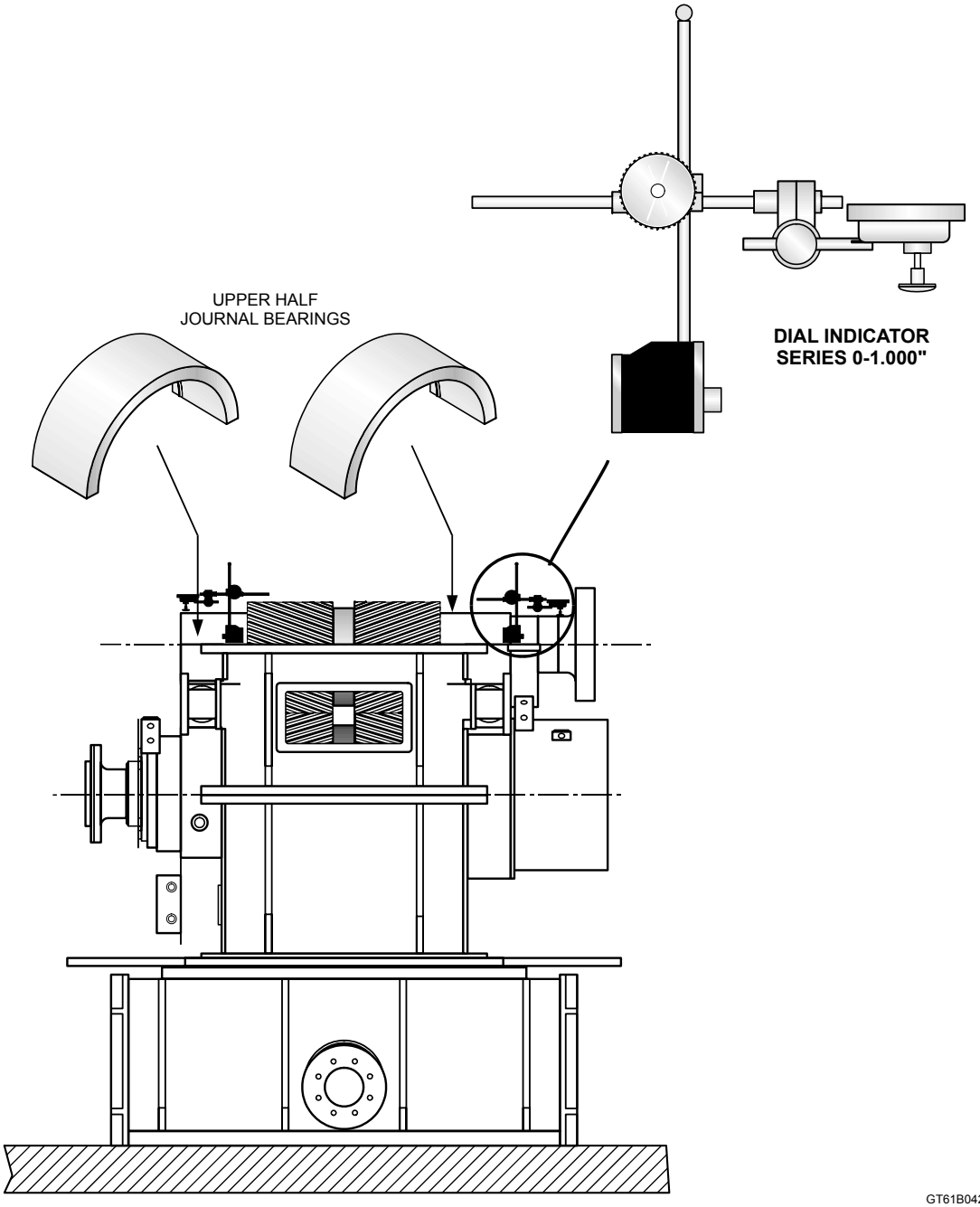


Figure 16. Measurement Arrangement for Upward Movement of Pinion Gear Shaft

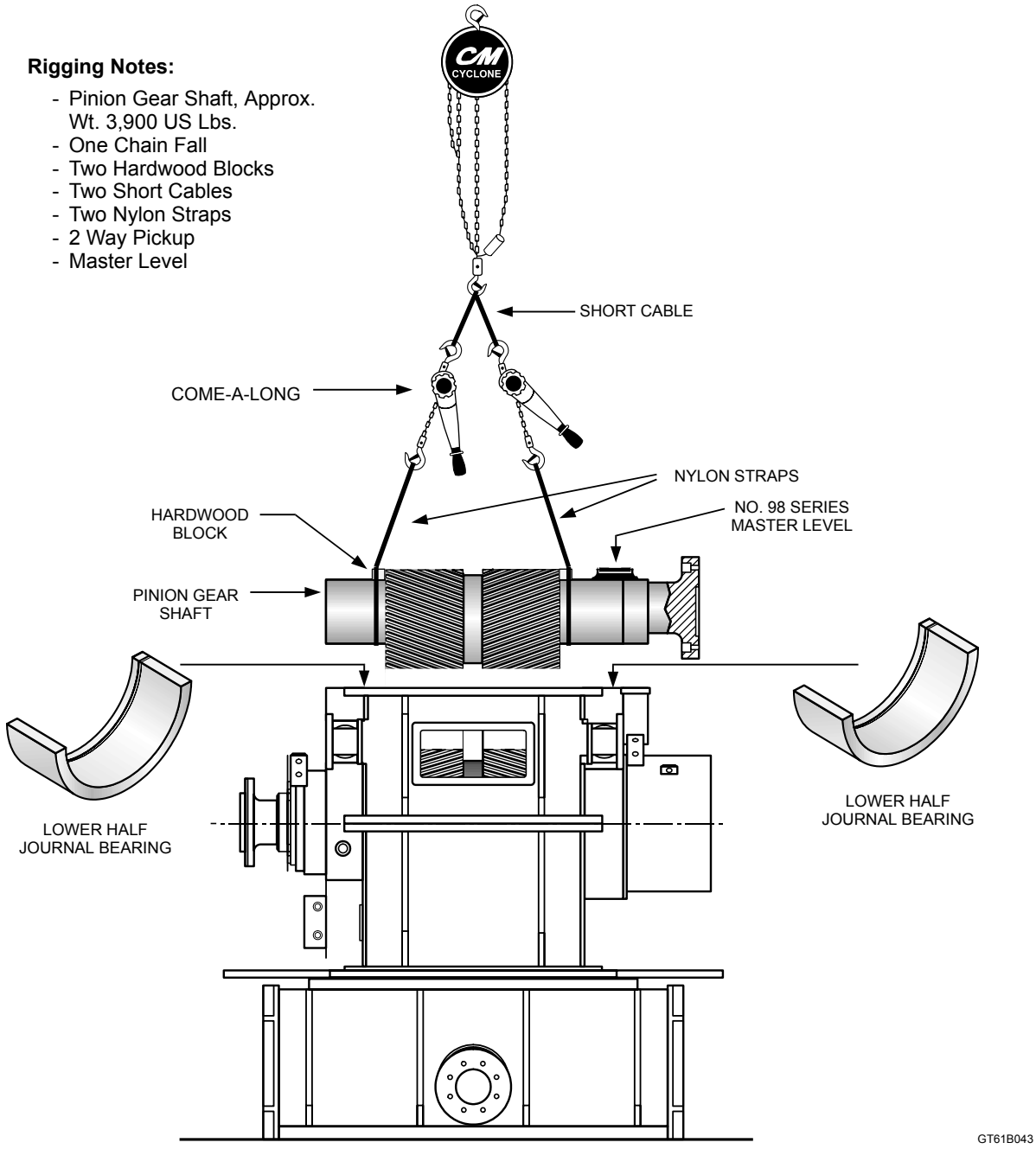


Figure 17. Pinion Gear Shaft Rigging Arrangement

5. Mount the master level on the gear shaft, and slowly adjust both come-a-longs until the gear shaft begins to move, then **stop**. Adjust the come-a-longs until the master level reads 0.000 inches.

Note: Care must be taken not to drop or damage the Master Level.

6. Slowly adjust the chain fall upward while keeping the Master level reading at 0.000 inches. **Stop movement if the shaft is not perfectly level**. The critical part of the lift is to avoid placing any stress on the gear teeth as they disengage.
7. When the pinion gear teeth are clear of bull gear teeth, lift the pinion gear shaft with the chain fall until it has been raised 10 inches out of the bearings. (See Fig. 17).
8. Lift the pinion gear shaft (with the overhead crane) away from the gear case and set in a wooden box for protection and storage. (See Fig. 18).
9. Roll out the lower half of both journal bearings. (See Fig. 17).
10. Remove bolts and dowels from gear box upper case horizontal joint, install four (4) guide pins, and use four (4) jack bolts to jack the case up approximately 1.000 inch.

    Rig (using two come-a-longs), level (using a Master level), and slowly lift the upper case with the chain fall until it is clear of the bull gear. The upper half casing may now removed from the work area with the maintenance crane. (See Figs. 19 and 20),
11. Attach a dial indicator at the generator end of gear box to measure to upward movement of the bull gear shaft. (See Fig. 21).
12. Rig the bull gear shaft (turbine end) with a chain fall connected to hook of the maintenance crane and a nylon strap around the gear shaft.

    While watching the dial indicator, slowly lift gear shaft with the chain fall until the indicator reads 0.005 inches. (See Fig. 21).
13. Roll the lower half of the journal bearing around to the top of the shaft, and remove. Inspect for damage and clean with soft rags. (See Fig. 21).
14. If the bearing is in good condition, re-assemble (**dry**), lower the chain fall and return gear shaft to rest in the bearing.
15. Rig the bull gear shaft (generator end) with a chain fall connected on the hook of the maintenance crane, and a nylon sling around the gear shaft.

    While monitoring the dial indicator, slowly lift gear shaft with the chain fall until the indicator reads 0.005 inches. (See Fig. 22).

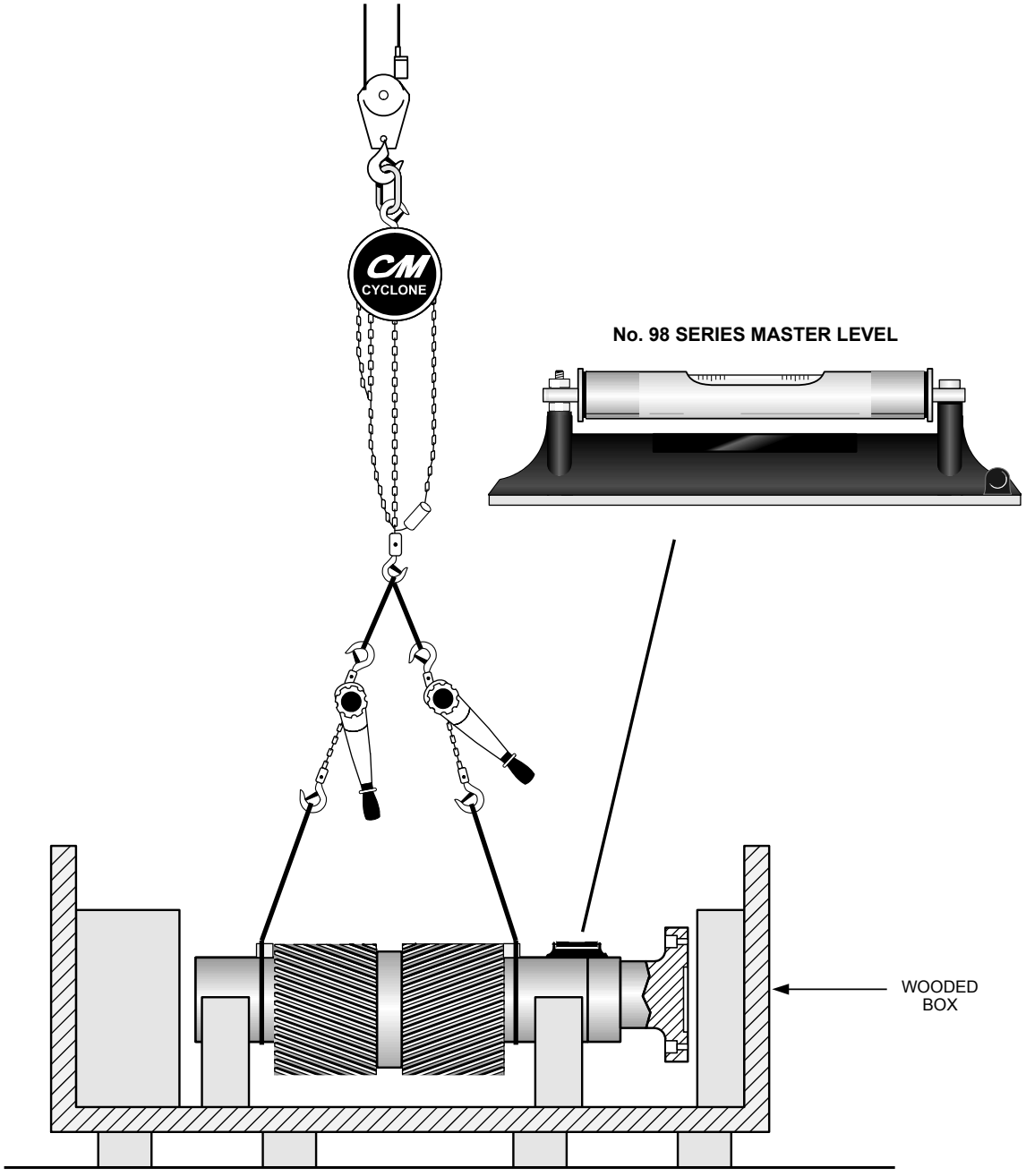


Figure 18. Rigging Arrangement for Pinion Gear Lift

Rigging Notes:

- 4 Way Pickup
- Upper Case, Approx. Wt.
- One Chain Fall
- Two 8 Ft. Cables
- 4 short cables
- 2-Come-a-longs
- Master Level

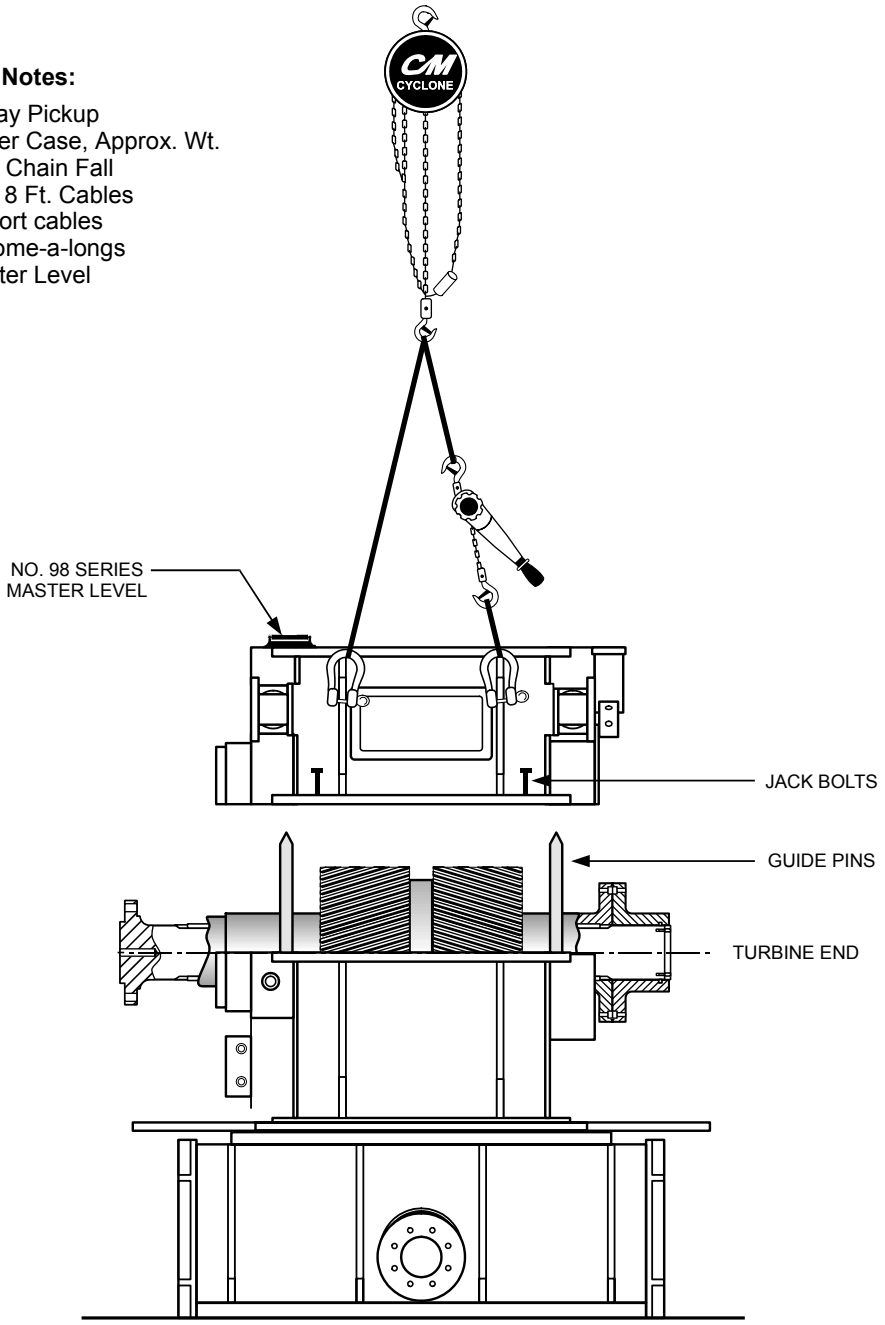
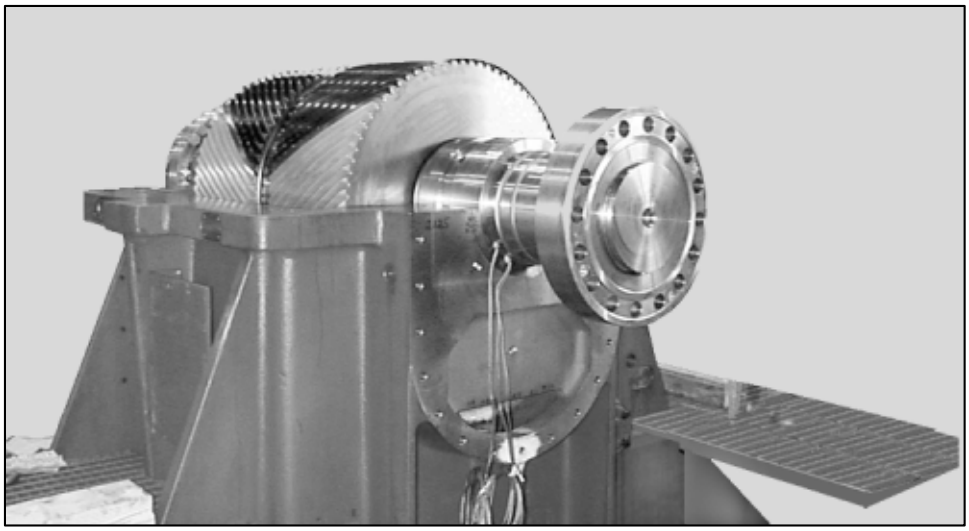
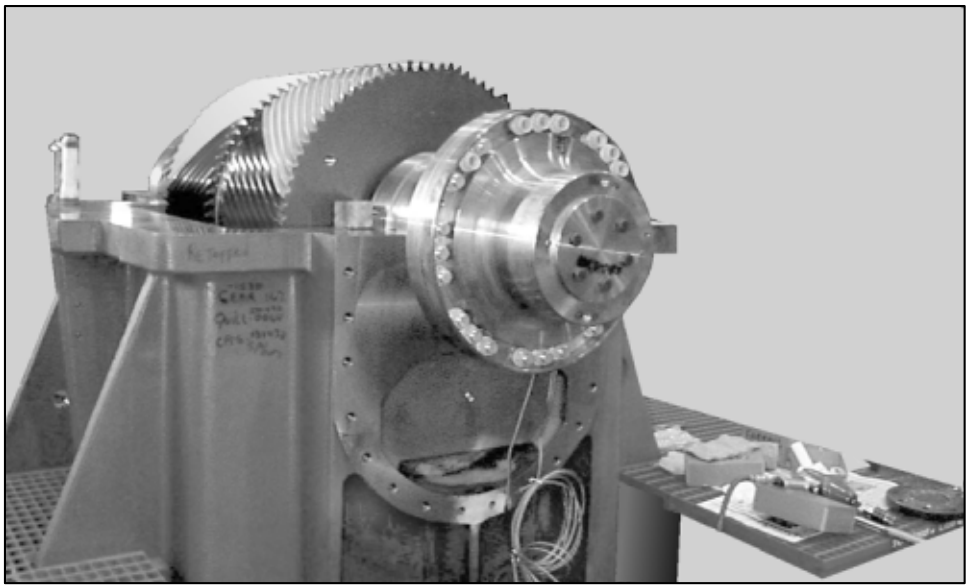


Figure 19. Upper Gear Case Rigging Arrangement



Generator End



Turbine End

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Figure 20. Load Gear with Upper Case Removed

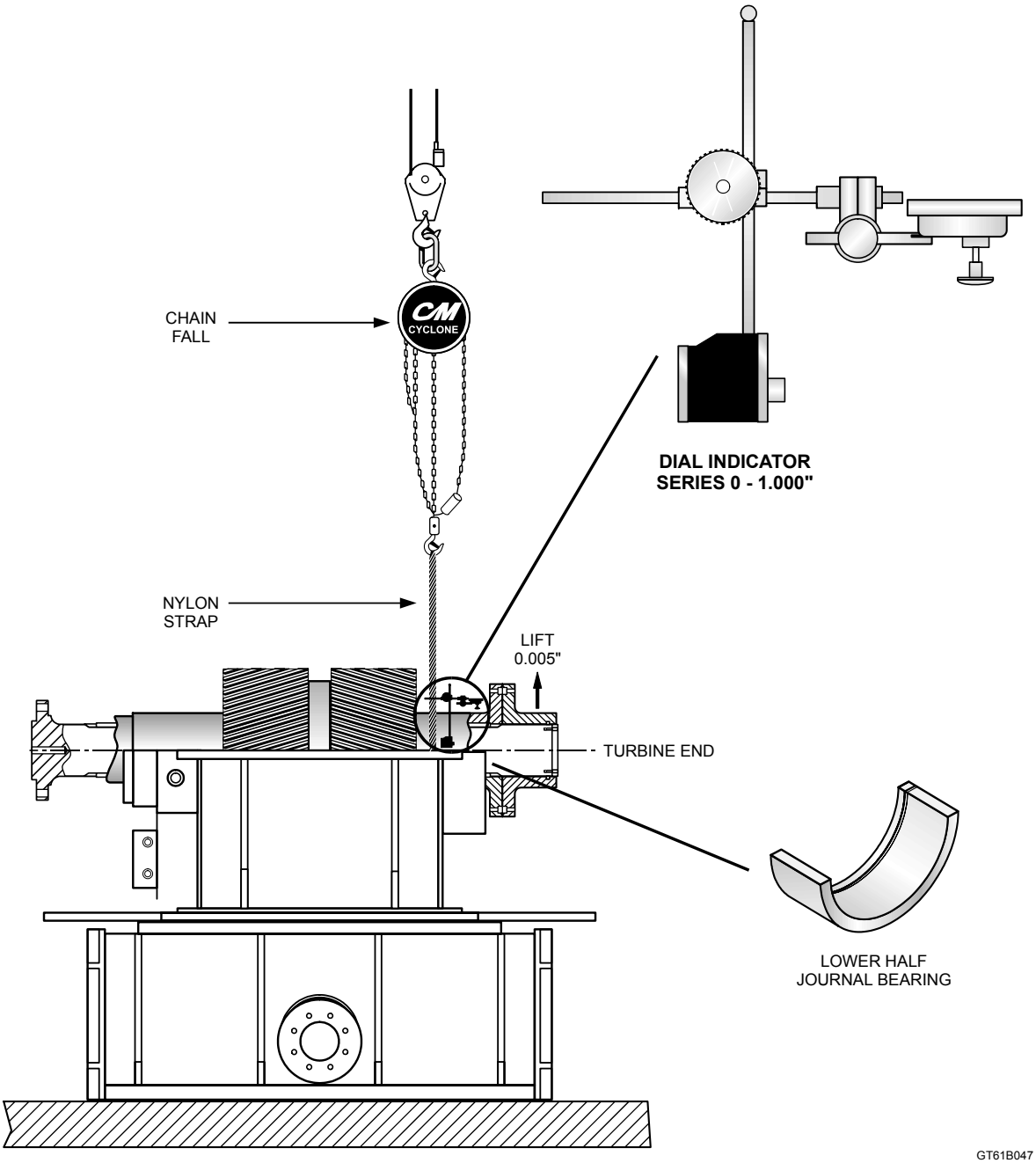


Figure 21. Turbine–end Lower Half Journal Bearing Removal

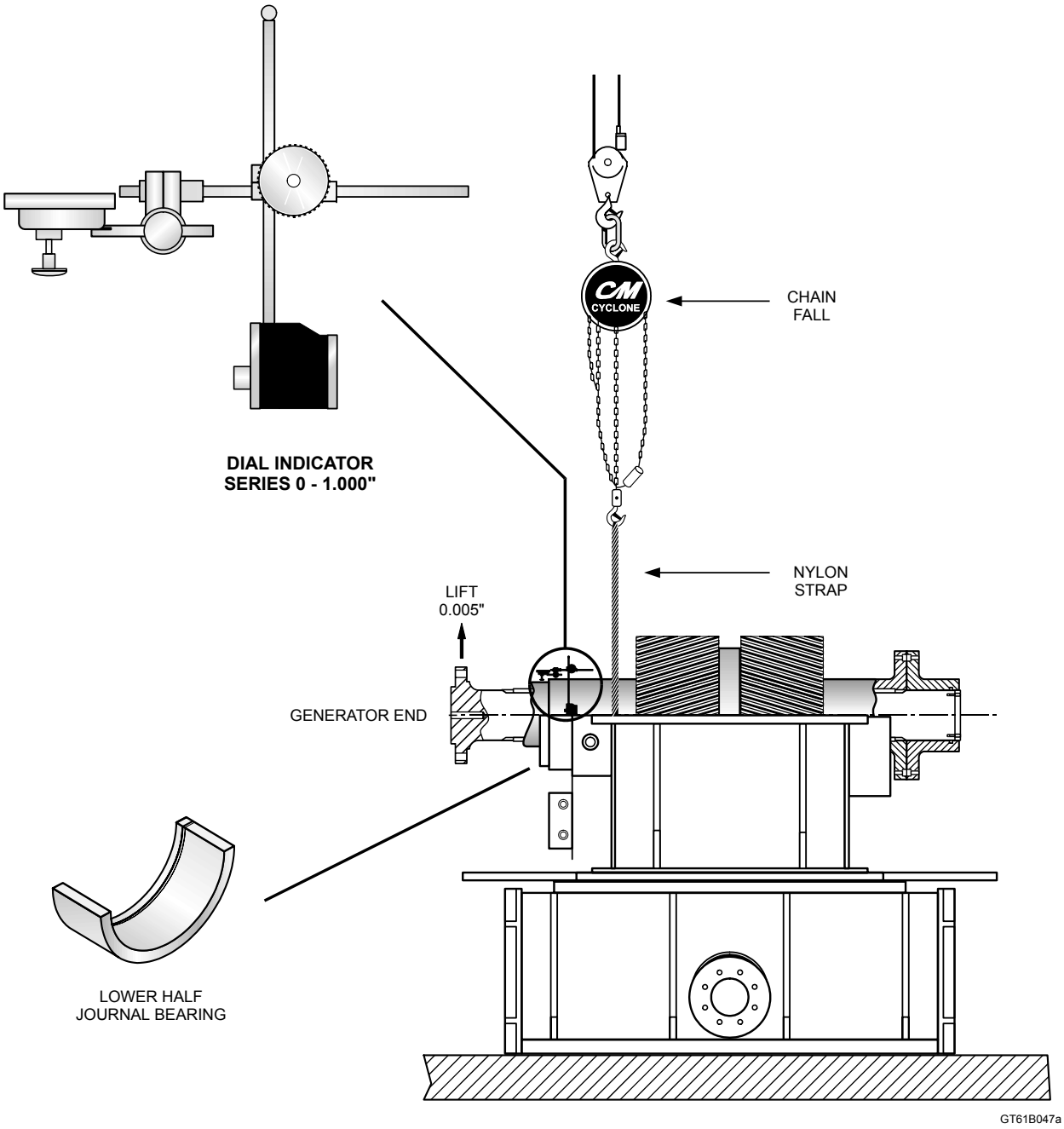


Figure 22. Generator–end Lower Half Journal Bearing Removal

16. Roll the lower half of the journal bearing around to the top of the shaft and remove. Inspect for damage, and clean with a soft rag. (See Fig. 22)
17. If the bearing is in good condition, wipe clean and re-assemble (**dry**), lower chain fall and return gear shaft to rest in the bearing.
18. Place a piece of lead fuse wire on top of both bull gear shaft journals (running the entire bearing length), and reassemble the upper half of both bearings. (See Fig. 23).
19. Re-assemble the gear box upper case (using the rigging shown in step 10 and See Fig. 19). Assemble bolting hardware and torque to vendor spec.
20. Repeat Step 10. — Removal of gear box upper casing.
21. Remove the upper half of both bull gear shaft bearings and carefully remove each piece of lead fuse wire, mic the thickness and record clearance reading. (See Fig. 24).

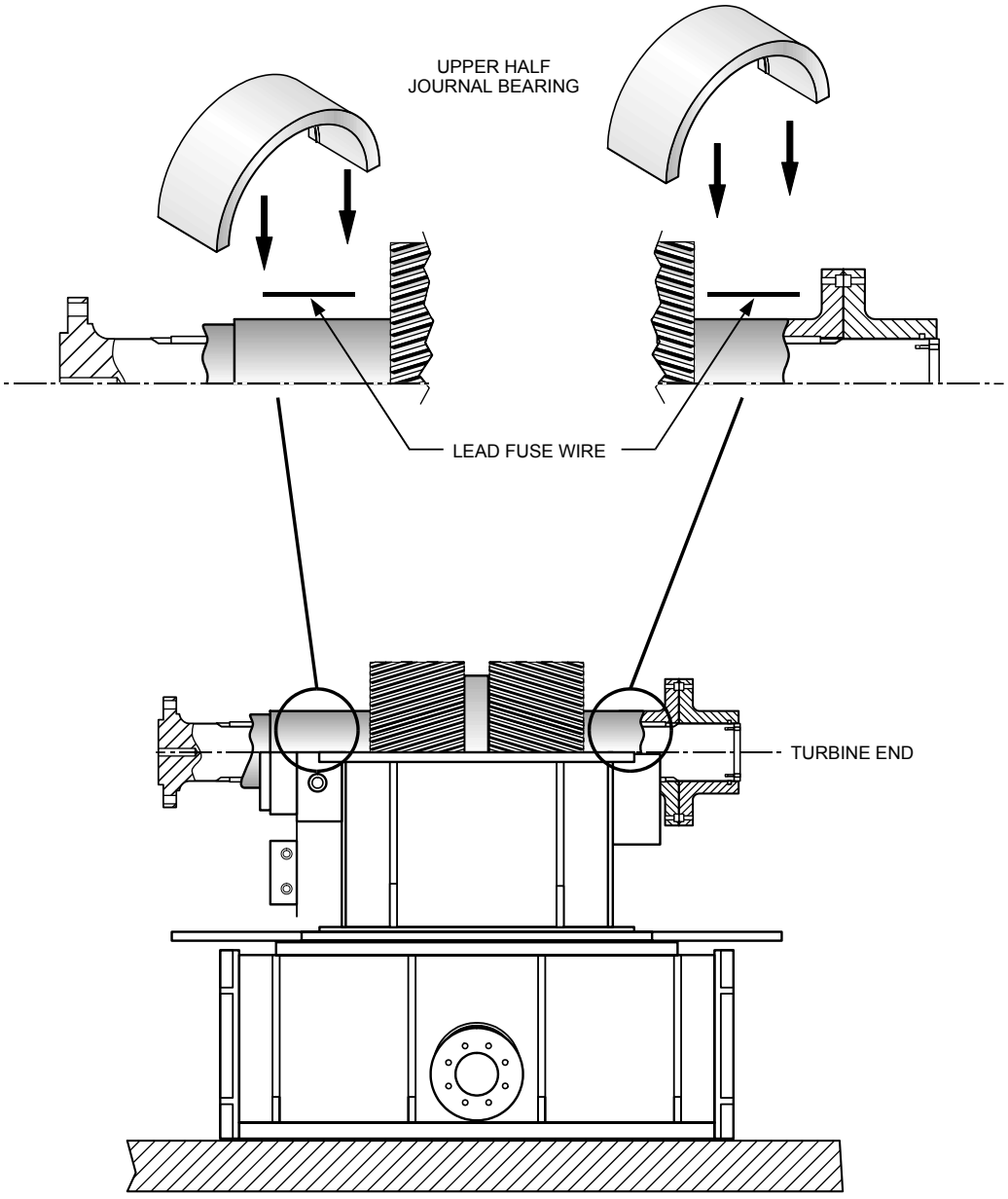
Note: Keep each piece of lead fuse wire as proof of clearances. Place them in a plastic bag and attach to inspection report.

22. Using a piece of crocus cloth, lightly clean the babbitt in the upper half of both bearings. This will remove any remaining specks of fuse wire.
23. Take a handful of turbine lube oil and rub it over both journals of the bull gear shaft. Set both bearings down over the journals making sure they are seated.

Note: Insure hands are clean before putting oil on bearings.

Load Gear Reassembly

1. Repeat step 19. — Reassemble the gear box upper casing. Apply a bead of red RTV to the horizontal joint of the lower gear case before setting the upper case. When the upper casing is installed, turn the bull gear shaft 360 degrees to lubricate the bearings.
2. Clean and assemble both lower journal bearing halves of the pinion gear shaft (**dry**) in the upper gear casing. (See Fig. 17).
3. With a Master Level (Model 98) mounted on the pinion gear shaft, rig and level the shaft by slowly adjusting both come-a-longs until the gear shaft begins to move. Then **stop,** and adjust each come-a-long until the master level reads 0.000 inches. (See Fig. 18).
4. Slowly adjust the chain fall up while keeping the master level reading 0.000 inches. **Stop** if the shaft is not perfectly level. (The gear teeth of the pinion gear will not mesh with the bull gear teeth, unless the shaft is level.) (See Fig. 18).



Note: Fuse wire must run the entire length of bearing surface.

Figure 23. Measuring Bearing Clearance with Fuse Wire

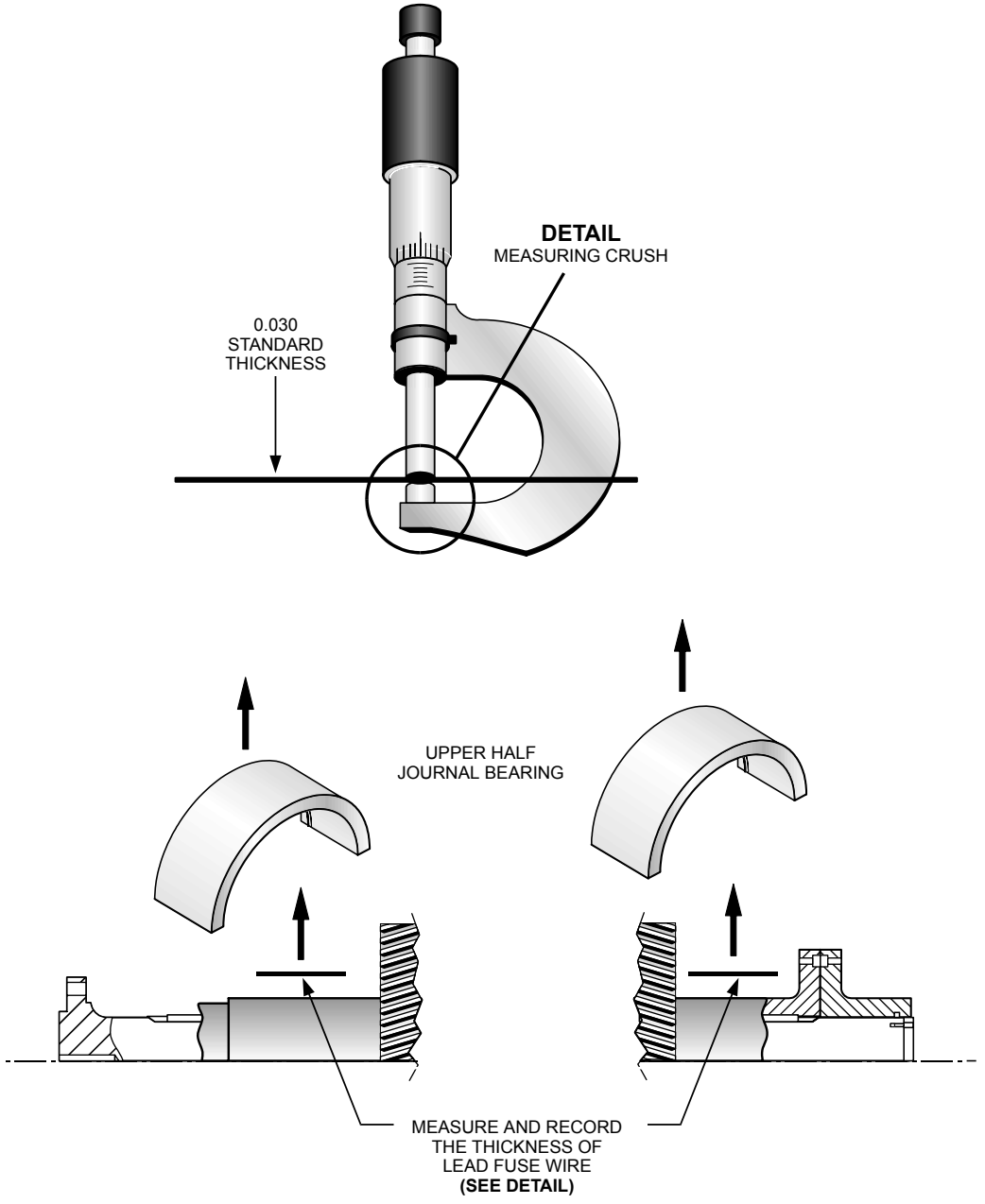


Figure 24. Reading Bearing Clearance with Crushed Fuse Wire

5. Move the maintenance crane into position (carrying the pinion gear shaft) over load gear. Lower crane hook until the load is approximately 10 inches over the upper gear case. (See Fig. 17).

CAUTION: Great care must be taken during this part of the re-assembly to assure that the match marks on both gears are lined up perfectly and the pinion gear teeth slide into mesh.

Note: The required crew normally consists of four millwrights; One to operate the chain fall, two to monitor the bearings, and one to monitor the gear teeth through the inspection cover hole as they engage.

6. Using the chain fall, slowly lower the pinion gear shaft down into lower halves of both bearings. (See Fig. 25).

7. Place a piece of lead fuse wire on top of both pinion gear shaft journals (running axially the entire length of bearing area). Reassemble the upper half of both bearings. (See Fig. 26).

8. Re-assemble gear box upper cover (using the rigging indicated Step 1, and shown in Fig. 15). Assemble bolting hardware and torque to the vendor specification.

9. Repeat Step 1. — Remove gear box upper cover.

10. Remove the upper half of both pinion gear shaft bearings and carefully remove each piece of lead fuse wire, mic the thickness and record clearance reading. (See Fig. 27).

Note: Keep the lead fuse wire as proof of clearances. Place each in a plastic bag and attach to the inspection report.

11. Using a piece of crocus cloth or Scotch Bright, lightly clean the babbitt in the upper half of both bearings. This will remove any remaining specks of fuse wire.

Note: The standard gas turbine gear bearing clearance is 0.0015 per inch of shaft size. If the clearance of a load gear bearing on any shaft exceeds 2X the normal clearance, both bearings must be replaced.

Continued operation with bearing clearences beyond the design limit is not recommended. The operation of gears with excessive clearance will result in damage to gear teeth. All bearaings must be replaced in pairs (e.g., both pinion bearings, both bull gear bearings.)

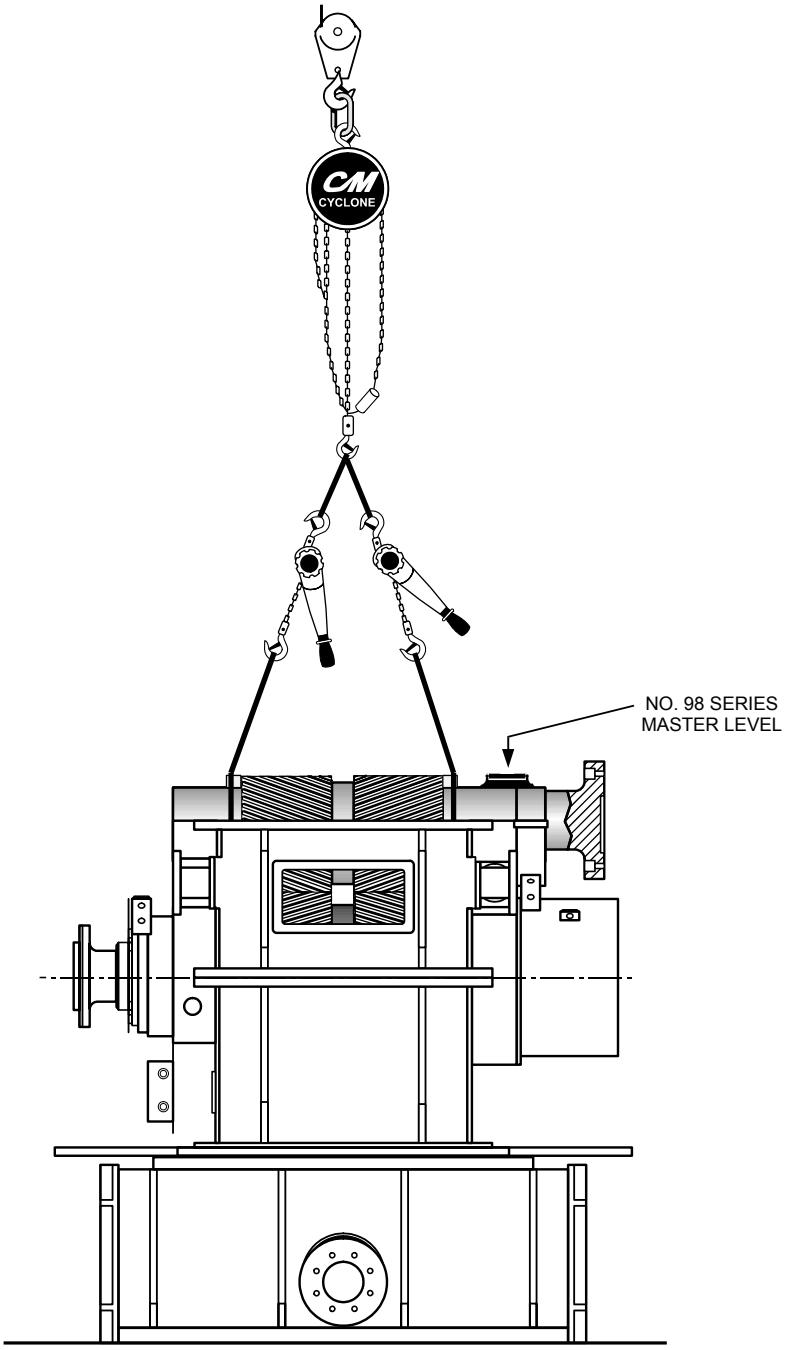


Figure 25. Reinstallation of Pinion Gear Shaft

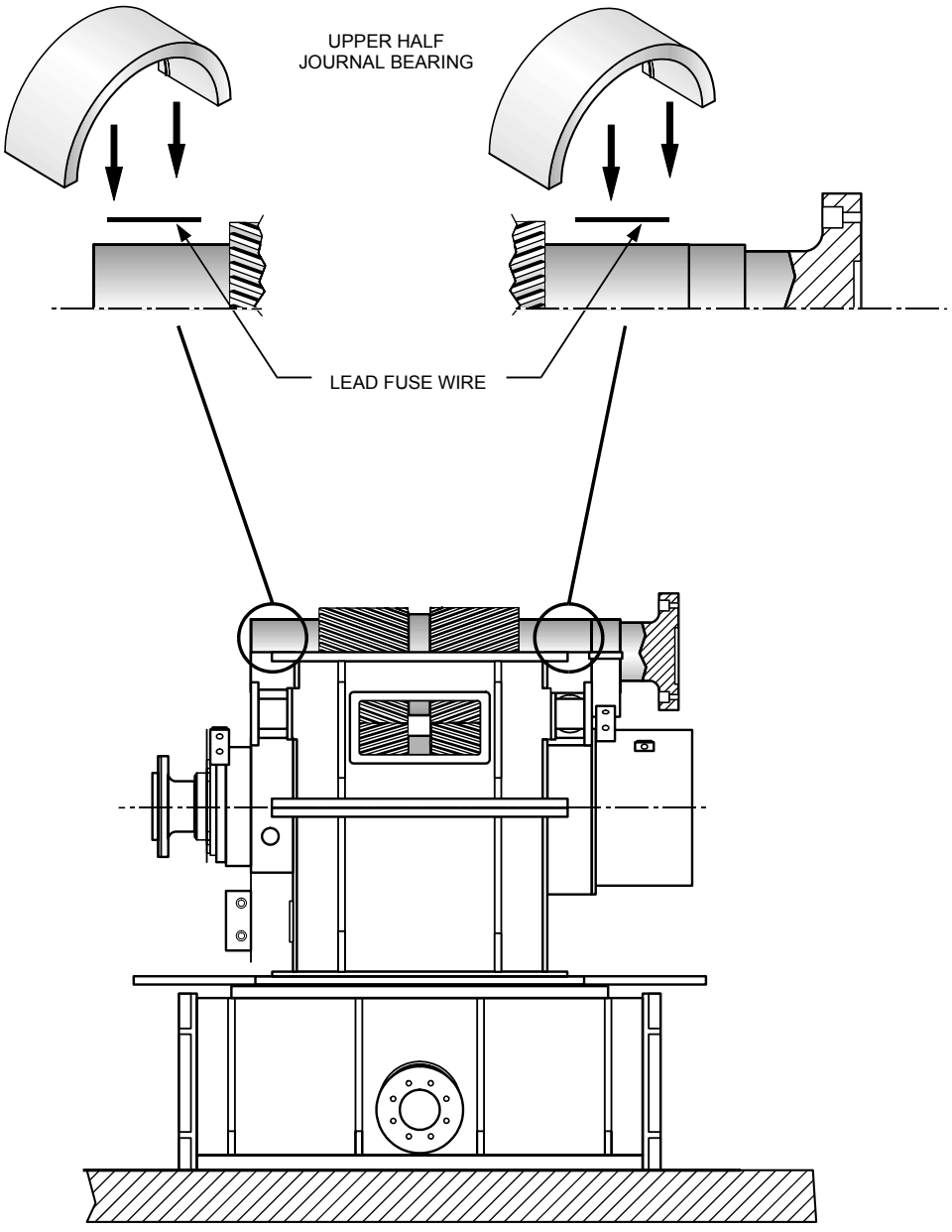


Figure 26. Measuring Pinion Gear Bearing Clearance with Fuse Wire

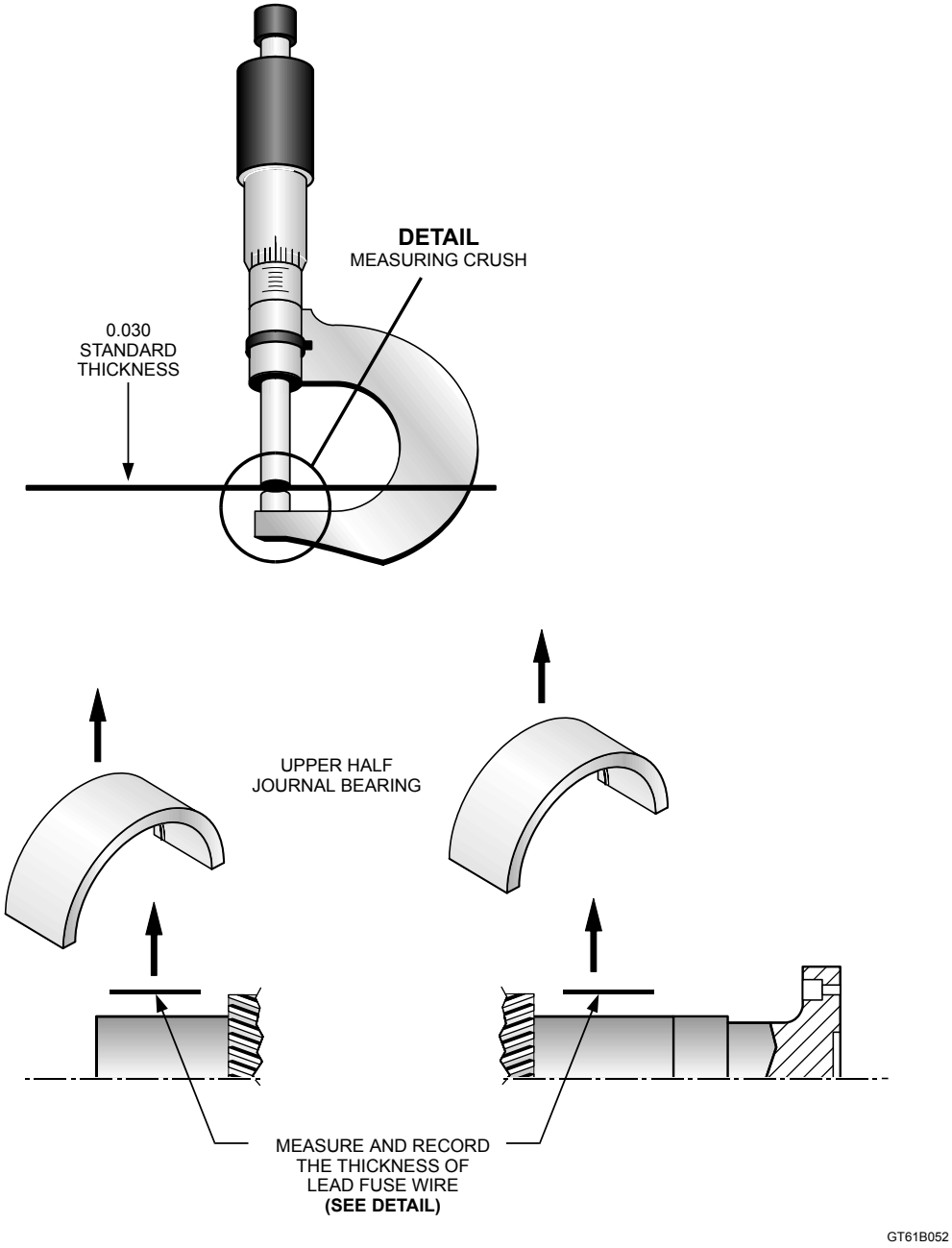


Figure 27. Reading Pinion Gear Bearing Clearance

12. Take a handful of turbine lube oil and rub your hand over both journals of the pinion gear shaft. Set the upper half of both bearings down over the journals making sure they are seated.

Note: Insure hands are clean before putting oil on bearings.

13. Repeat Step 8 (above) — Apply a bead of red RTV to the horizontal joint of the upper gear case before setting cover. Assemble bolting hardware and torque to vendor specification. Then turn the pinion gear shaft 360 degrees to lubricate bearings.

Check Load Gear Alignment

Generator End
Repeat Steps 1 through 6 in the generator part of the Load Gear Disassembly And Alignment Readings section of this article. (Including the instructions in Figs. 8 through 13).

WARNING: The centering ring must be removed from the generator coupling end after alignment is complete and prior to turbine operation. Failure to remove the centering ring will cause excessive vibration and may cause bearing failure and gear damage.

Gas Turbine End
Repeat Steps 1 and 2 in the gas turbine part of the Load Gear Disassembly And Alignment Readings section of this article. (See Fig. 14)

Note: Refer to Generator Alignment section Steps 3 through 6. Record "as-left" face and rim readings.

Reassemble Couplings

Generator
Refer to EM 5315 (Installation Procedure for MS6001B Gas Turbine – Generator Unit) for the generator/load gear coupling instructions.

Gas Turbine
Refer to EM 5316 (MS6001B Gas Turbine – Load Coupling Installation) for the as turbine/load gear load coupling instructions.